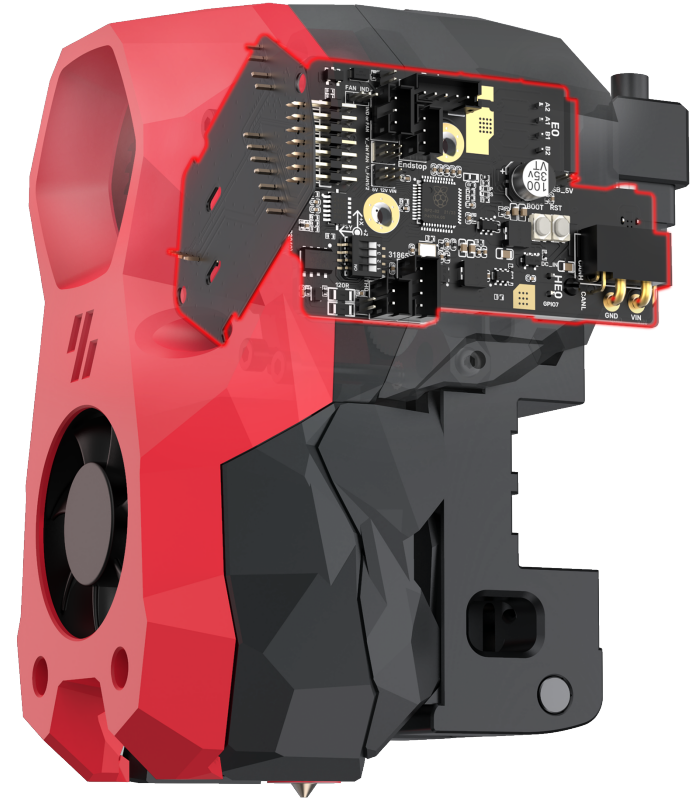
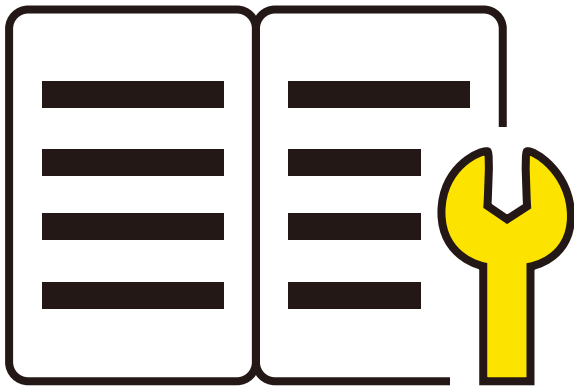




EBB SB2209 CAN V1.0 (RP2040)

BUILD GUIDE

VERSION 2023-11-21



Thanks to **CHAOTICLAB** for providing guidance on Voron's official style build guide, and designing the printed part model for EBB SB2209 CAN V1.0 (RP2040).

- Highlighted in blue are included in this EBB SB2209 CAN V1.0 (RP2040).
- Highlighted in red are other accessories of the Voron StealthBurner, which are not included in this EBB SB2209 CAN V1.0 (RP2040) and will need to be purchased by yourself.

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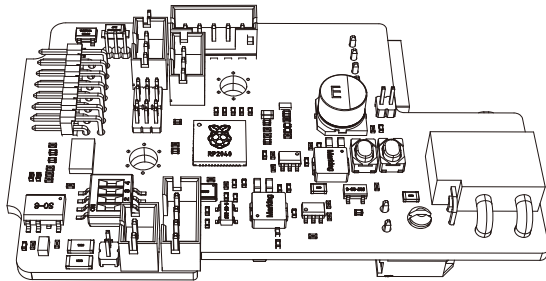
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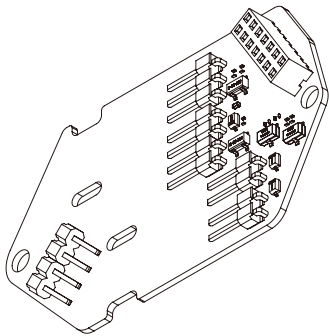
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PACKING LIST

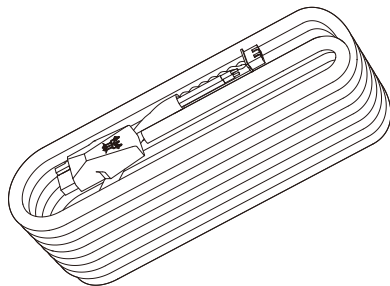
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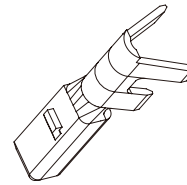
BIGTREETECH EBB SB2209 CAN V1.0 (RP2040) 1pc



BIGTREETECH EBB SB0000 CAN V1.0 1pc

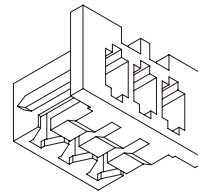


CAN bus Cable 1pc



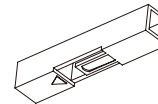
Crimp Terminal

2.0 mm pitch	30pcs
3.0 mm pitch	5pcs
2.54 mm pitch	20pcs



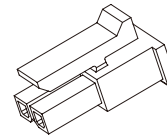
2.0 mm Pitch Connector Housing

2Way	1pc
3Way	2pcs
4Way	1pc
5Way	1pc



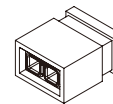
2.54mm Pitch Connector Housing

1Way	15pcs
------	-------



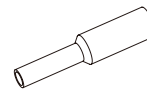
3.0 mm Pitch Connector Housing

2Way	1pc
------	-----



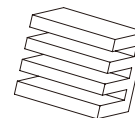
Jumper

15pcs



Insulated Wire Ferrule

2pcs



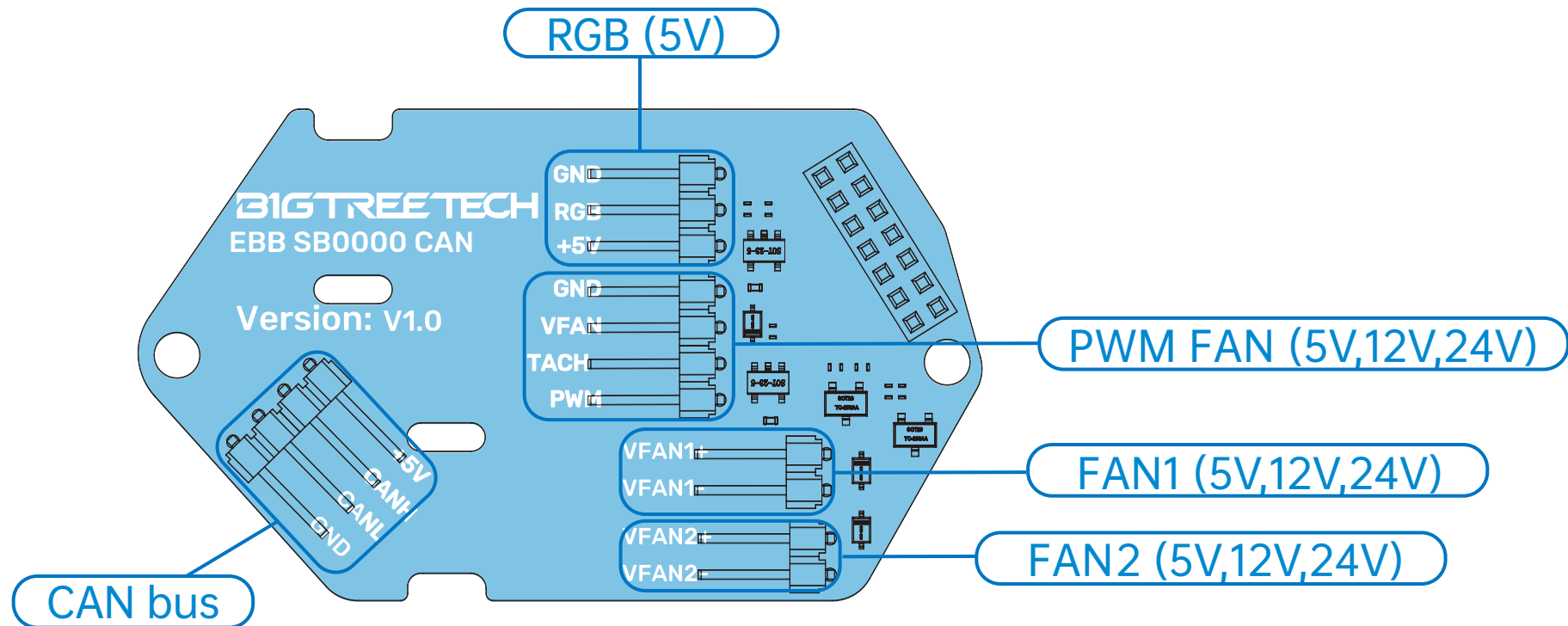
Heat Sink

1pc



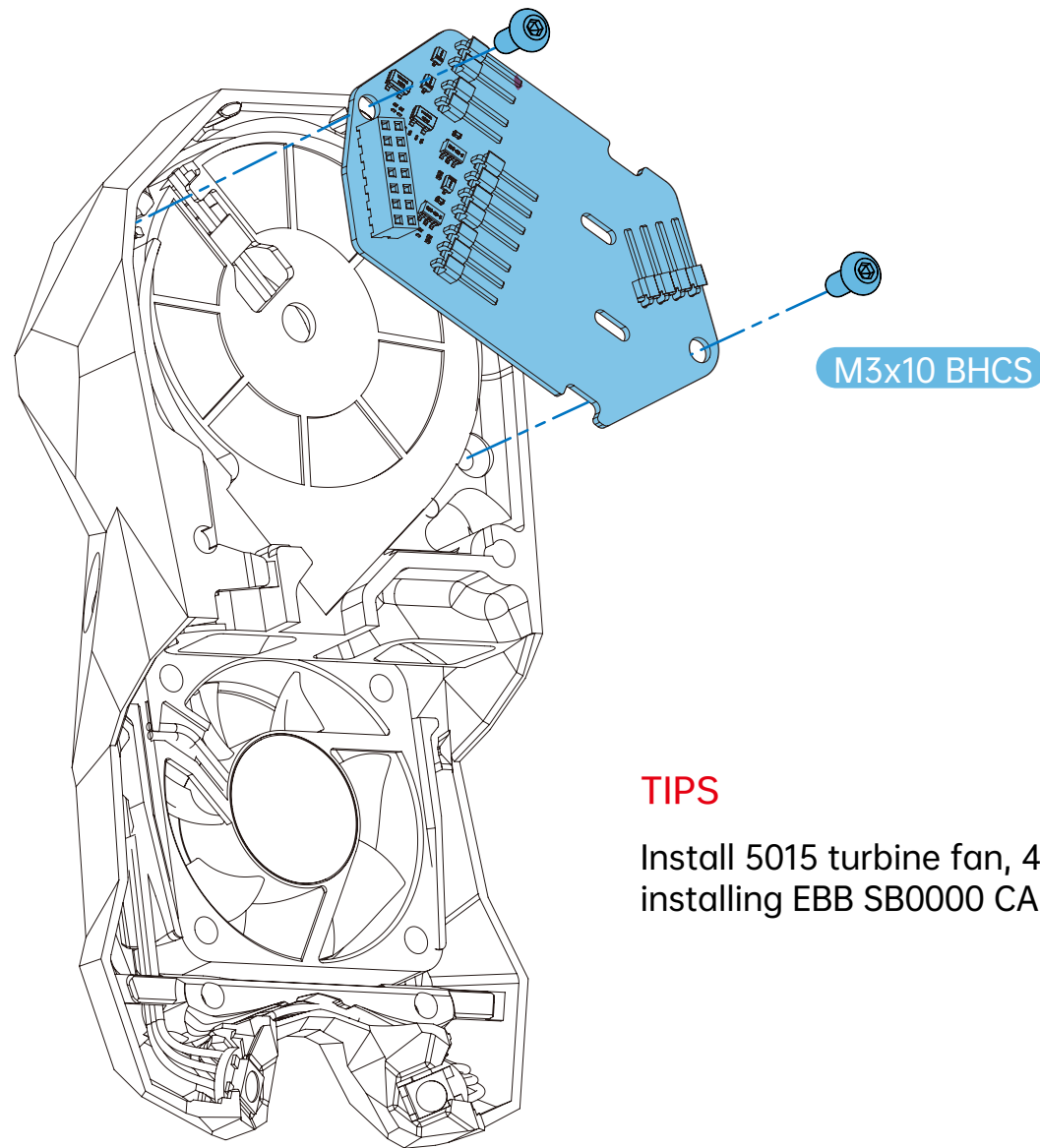
M3 x 10 Button Head Cap Screw (BHCS)

4pcs



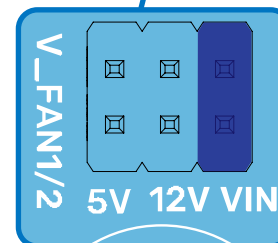
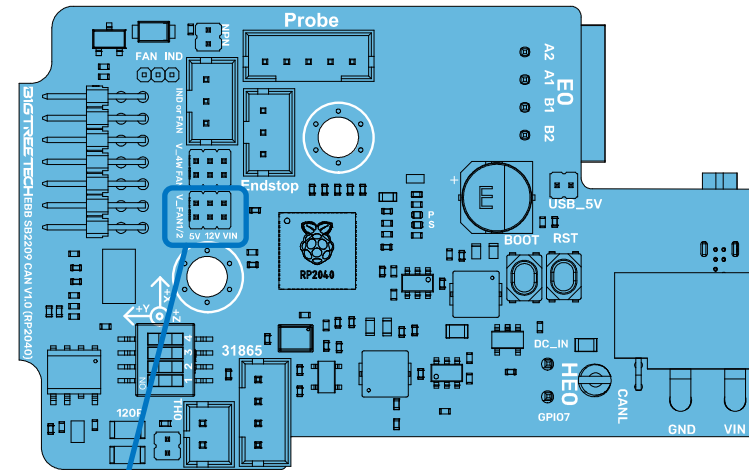
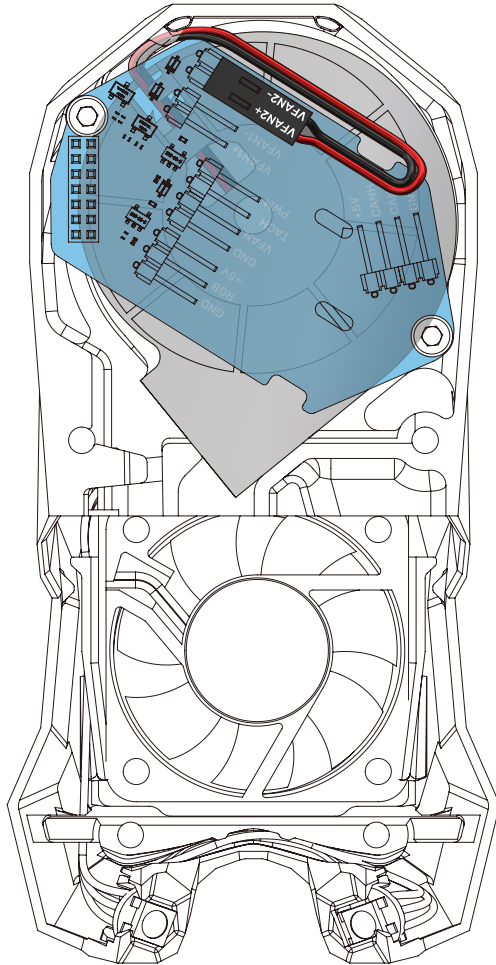
WIRE TERMINAL

EBB SB0000 CAN v1.0 use 2.54 pitch DuPont connector. Therefore, all fans and LED lights on StealthBurner Body should be 2.54 pitch DuPont connector.

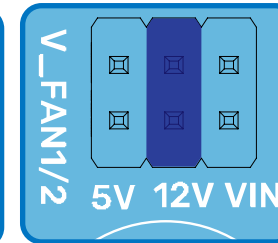


TIPS

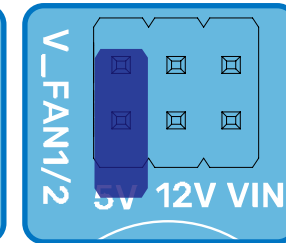
Install 5015 turbine fan, 4010 axial fan, and LED lamp before installing EBB SB0000 CAN v1.0.



Fan Voltage: 24V



Fan Voltage:12V

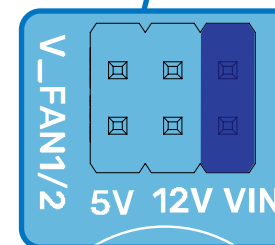
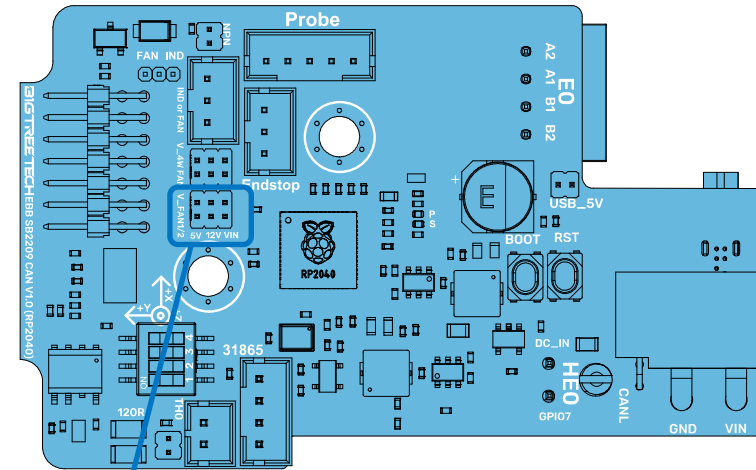
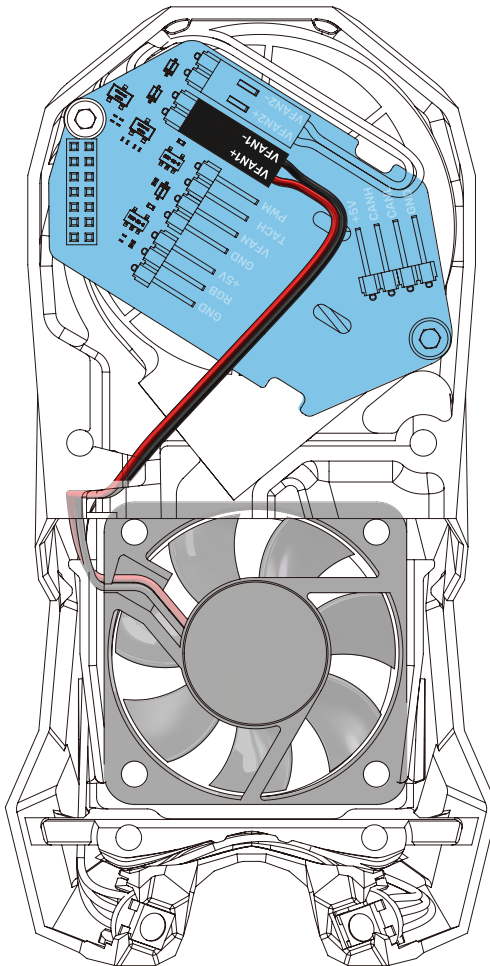


Fan Voltage: 5V

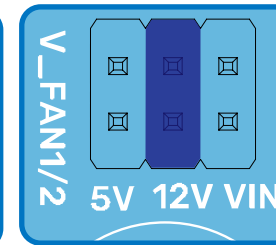
FAN VOLTAGE

All fans on the EBB SB0000 CAN v1.0 support multi voltage selection, as shown in the figure, and jumper caps are inserted or removed according to the actual voltage used.

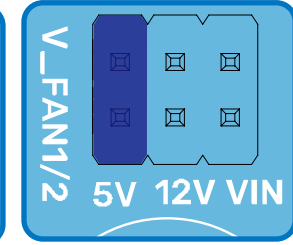
Note: The voltage for both the turbine fan and axial fan will be simultaneously set to 5/12/24V.



Fan Voltage: 24V



Fan Voltage:12V

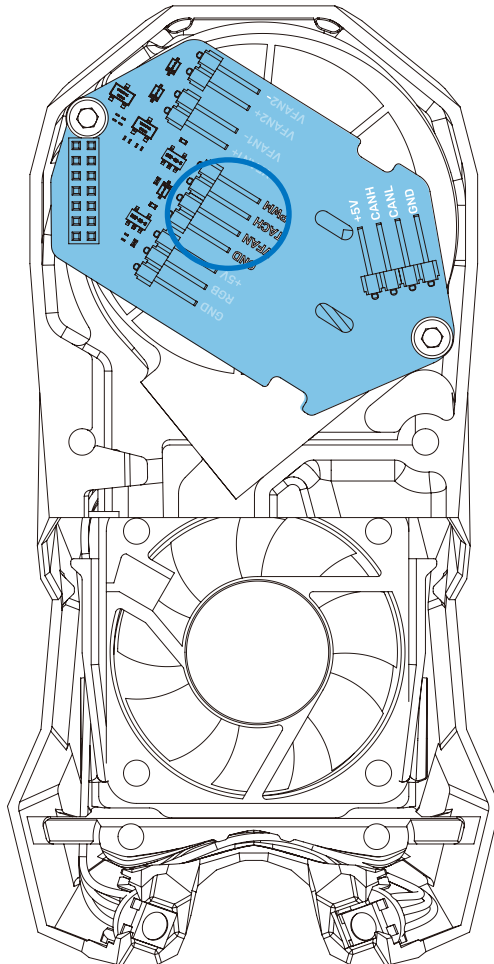


Fan Voltage: 5V

FAN VOLTAGE

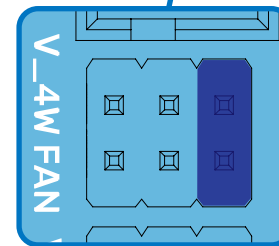
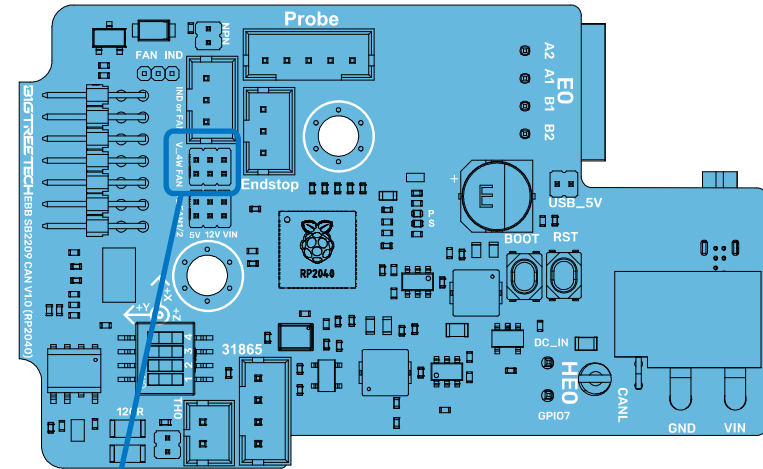
All fans on the EBB SB0000 CAN v1.0 support multi voltage selection, as shown in the figure, and jumper caps are inserted or removed according to the actual voltage used.

Note: The voltage for both the turbine fan and axial fan will be simultaneously set to 5/12/24V.

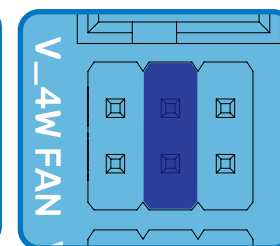


PWM FAN

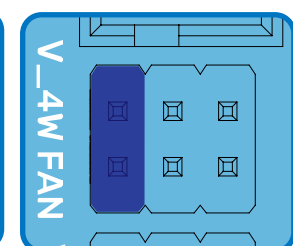
You can also use PWM fan on EBB SB0000 CAN v1.0. As shown in the figure, it contains a 4 pins PWM fan header (2.54 pitch and 4 pins DuPont connector).



Fan Voltage: 24V



Fan Voltage:12V



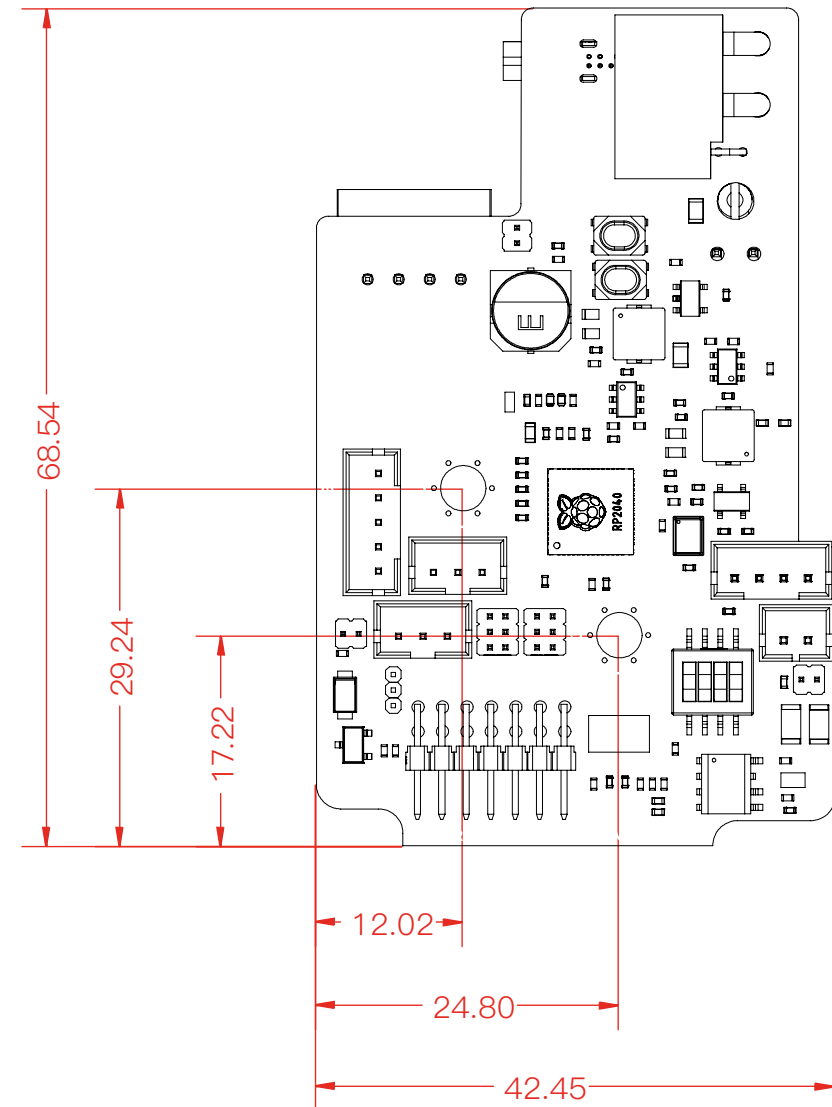
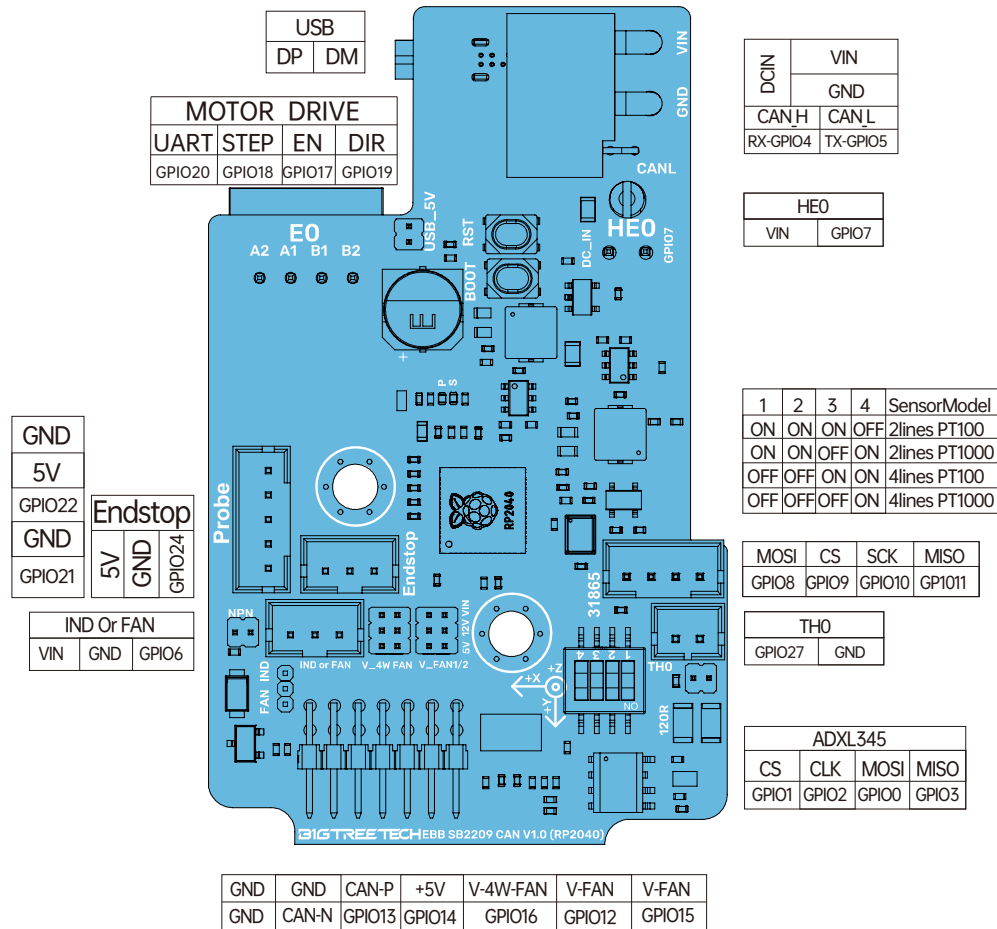
Fan Voltage: 5V

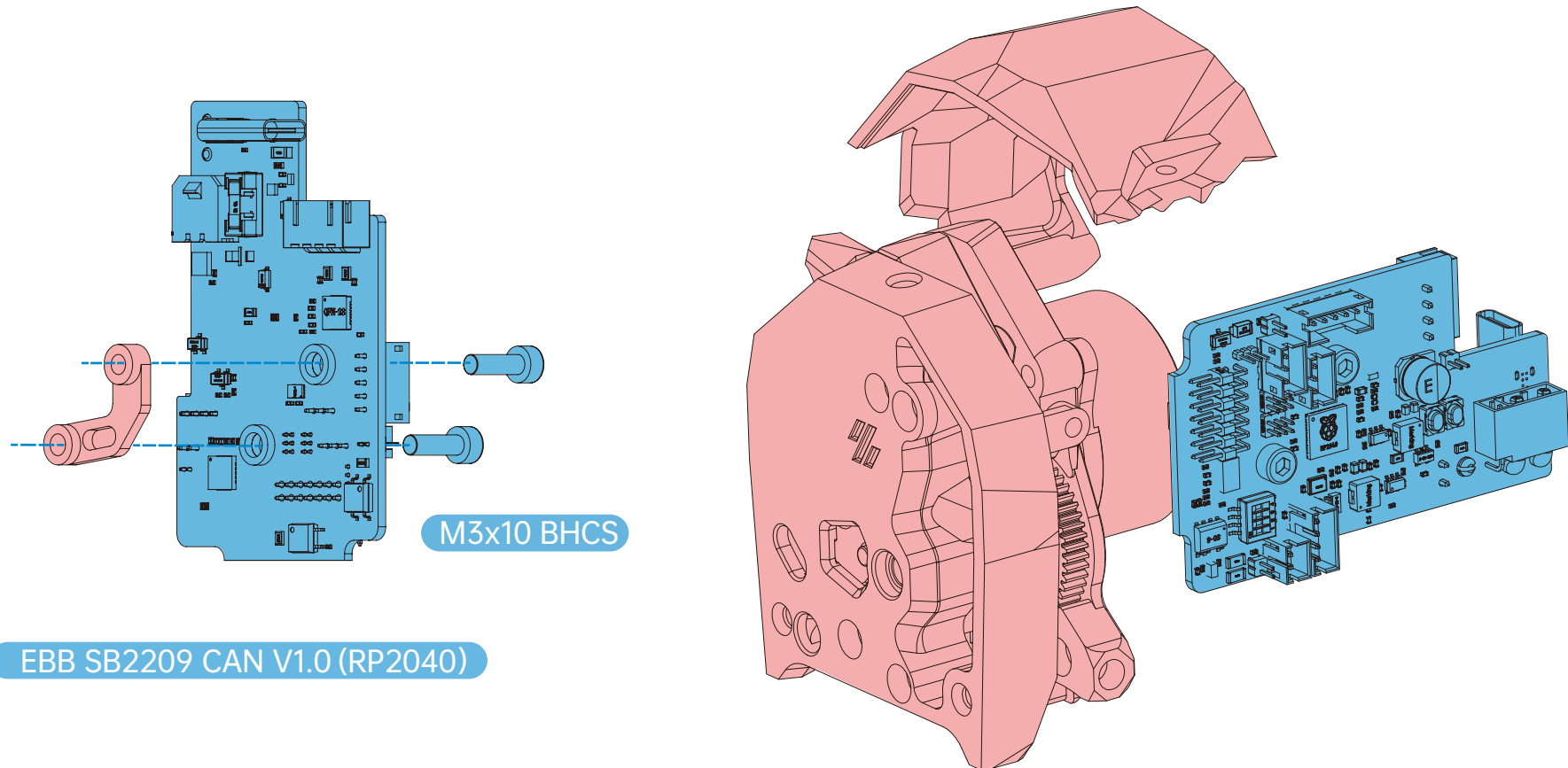
FAN VOLTAGE

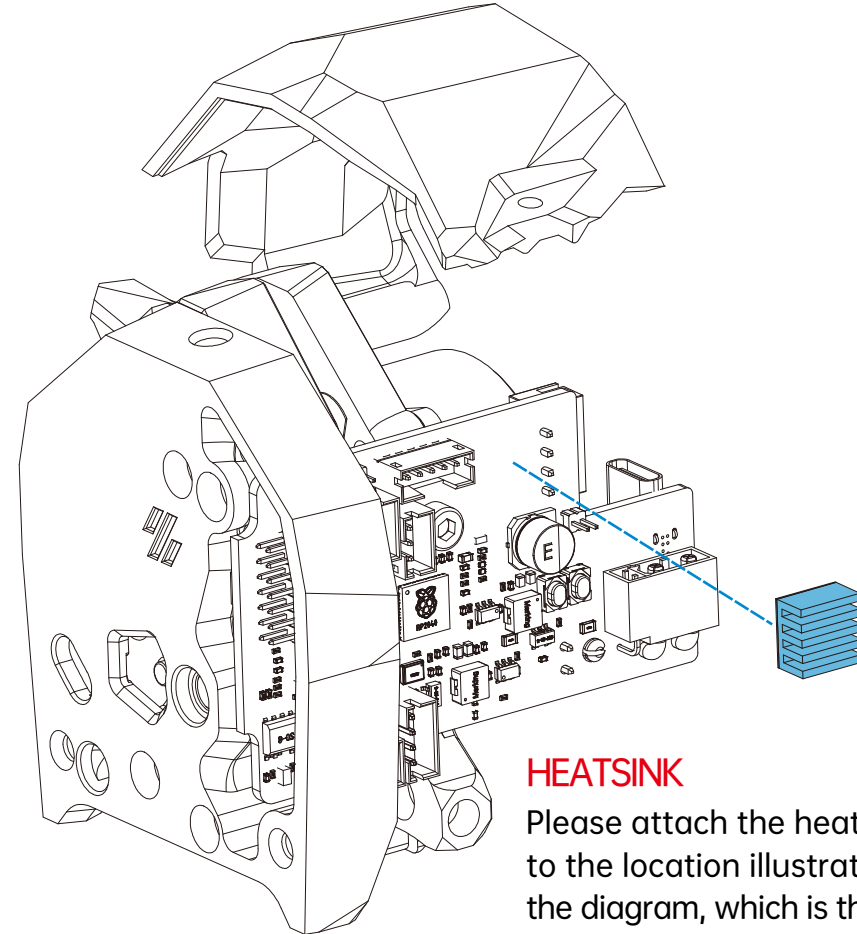
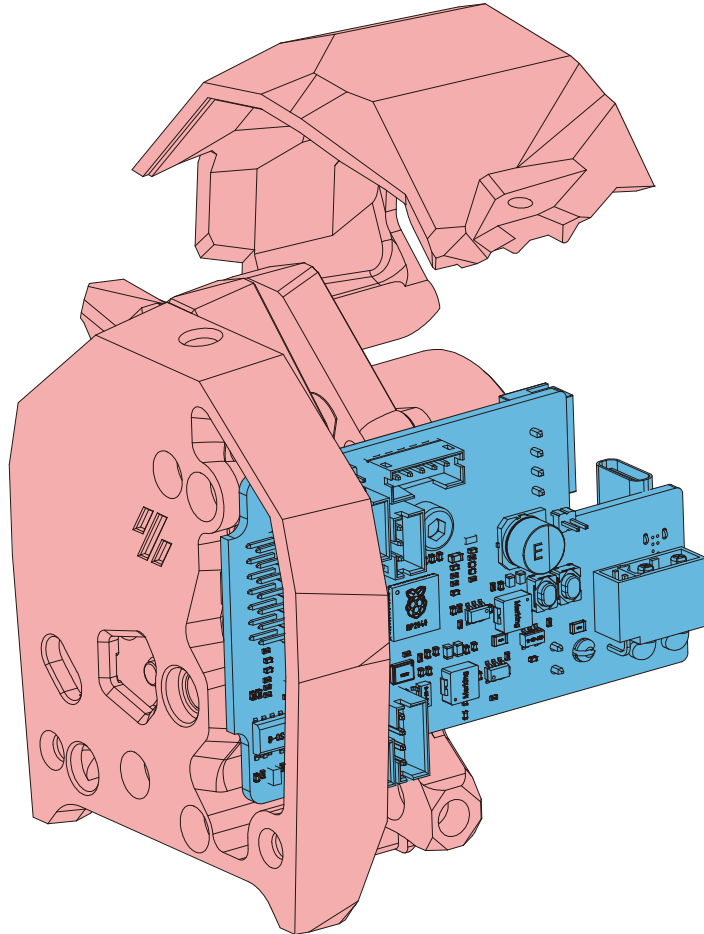
All fans on the EBB SB0000 CAN v1.0 support multi voltage selection, as shown in the figure, and jumper caps are inserted or removed according to the actual voltage used.

EBB SB2209 CAN V1.0 (RP2040) OVERVIEW

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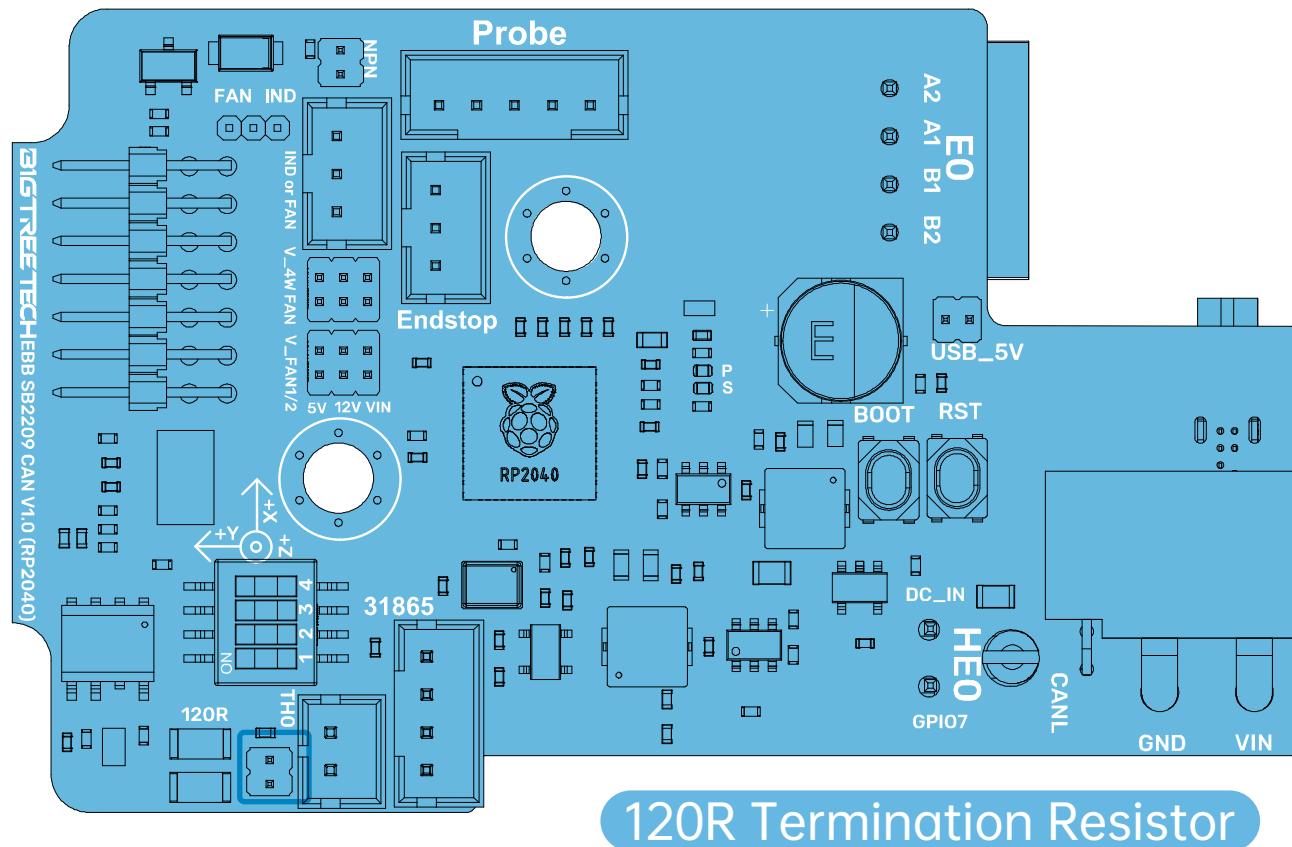


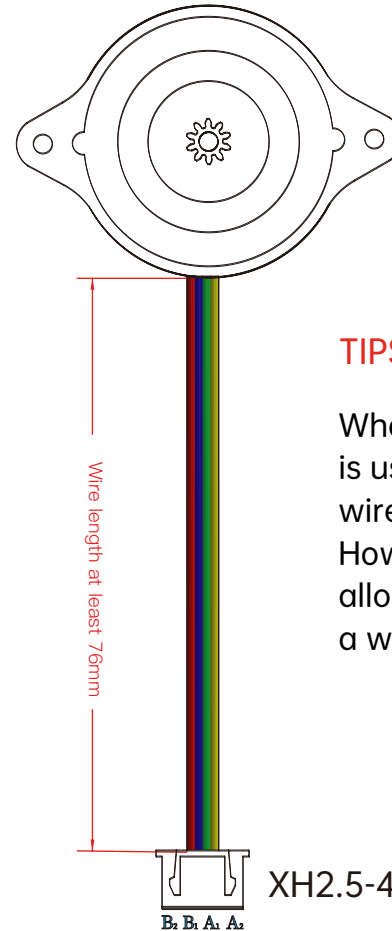
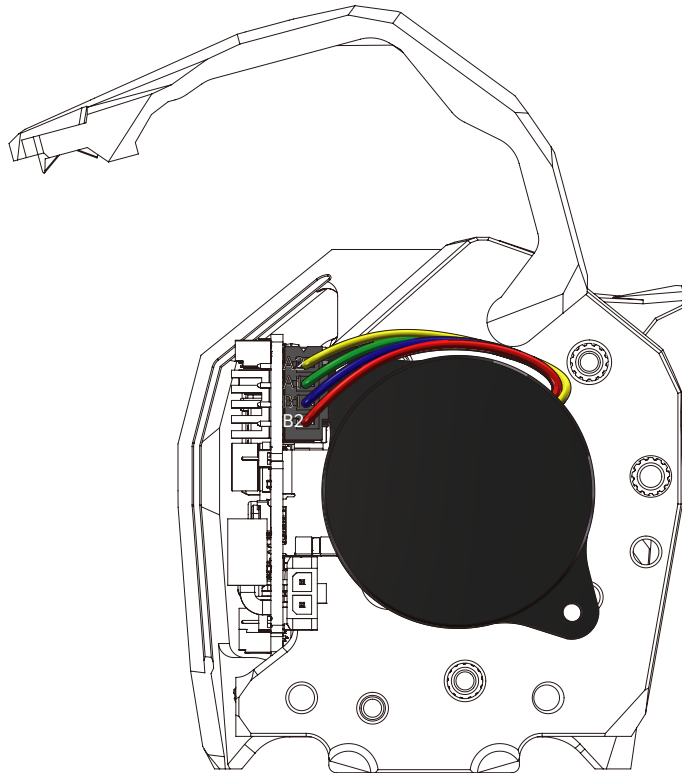


HEATSINK

Please attach the heatsink to the location illustrated on the diagram, which is the back of the drive chip.

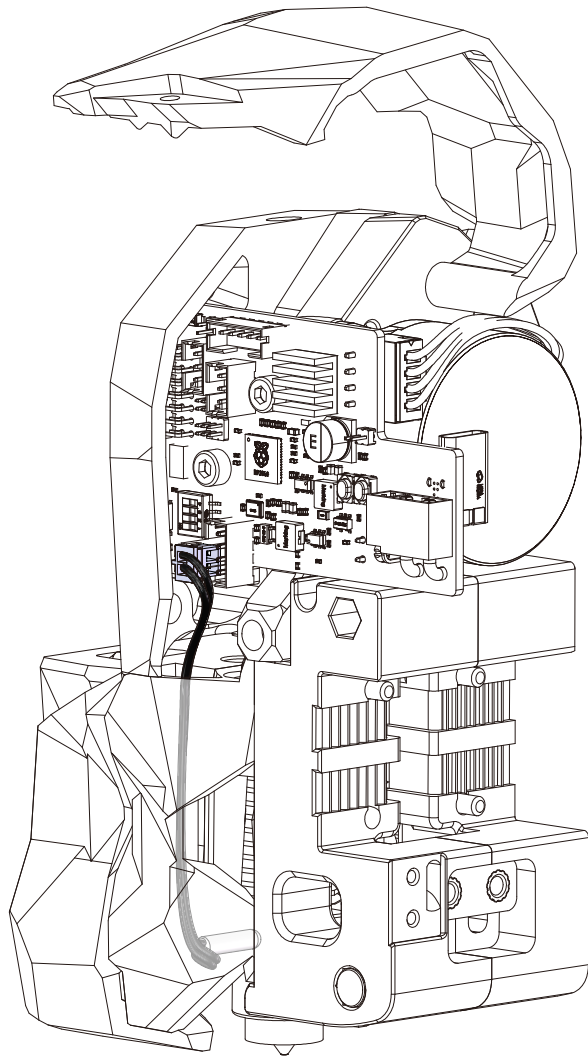
When the EBB SB2209 CAN V1.0 (RP2040) device uses CAN bus communication, if it is the final device in the CAN bus chain, you must plug a jumper at the 120R position.





TIPS

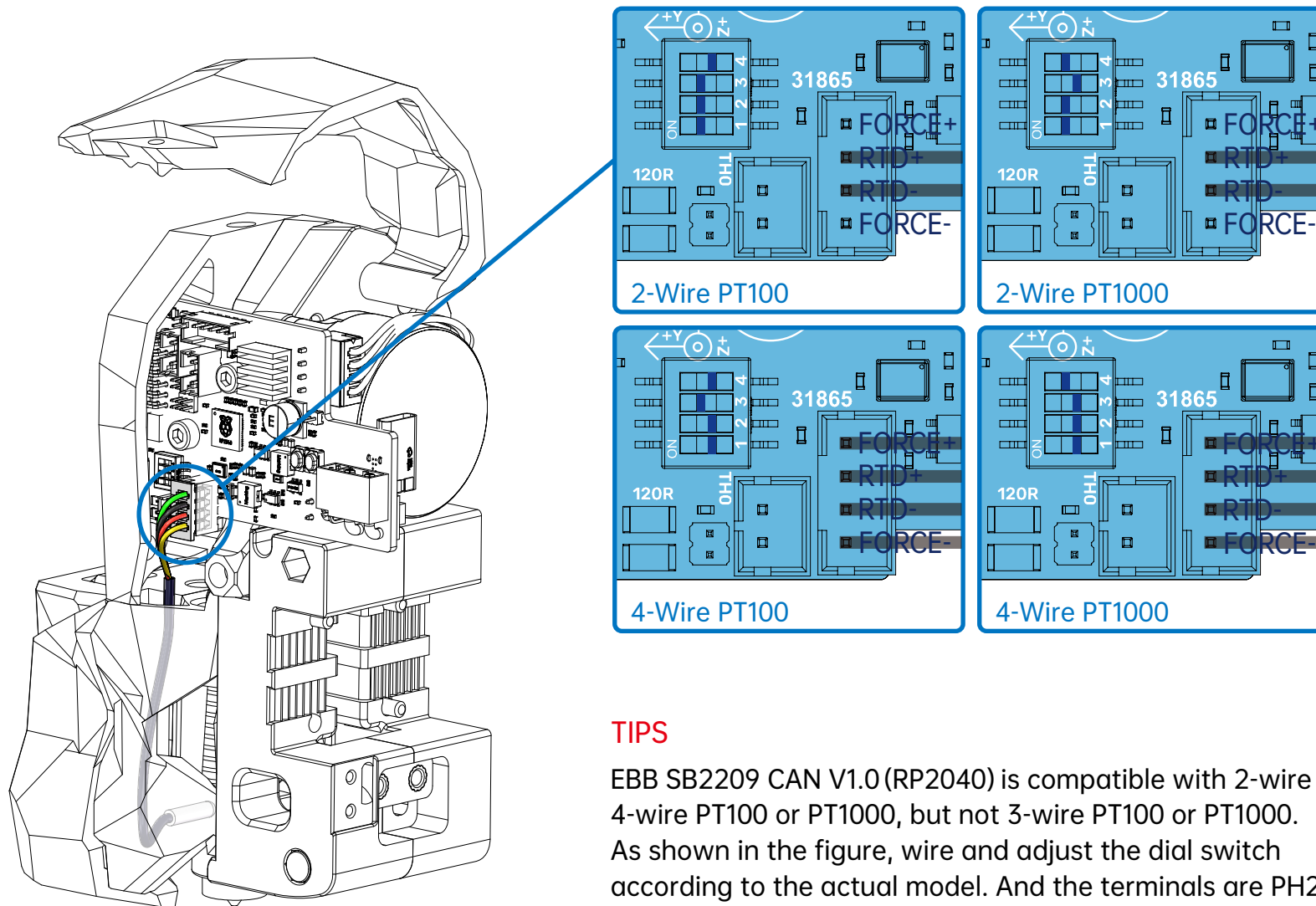
When EBB SB2209 CAN V1.0 (RP2040) is used on StealthBurner, the minimum wire length of the motor is 76mm. However, it is recommended to leave some allowance for wiring and assembly, such as a wire length of 100mm or 110mm.



PH2.0-2PIN

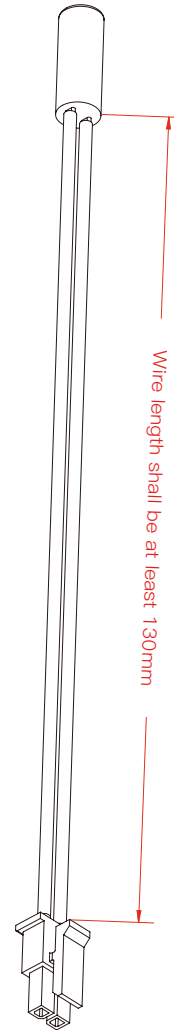
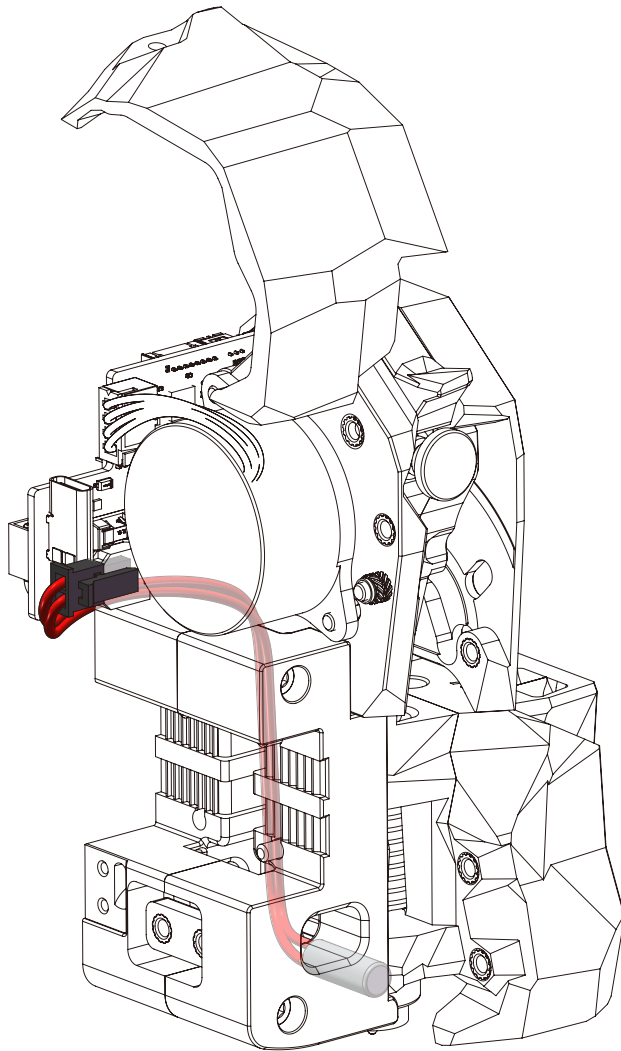
TIPS

When EBB SB2209 CAN V1.0 (RP2040) is used on StealthBurner, the minimum wire length of the thermistor is 103mm. However, it is recommended to leave some allowance for wiring and assembly, such as a wire length of 120mm or 130mm.



TIPS

EBB SB2209 CAN V1.0 (RP2040) is compatible with 2-wire and 4-wire PT100 or PT1000, but not 3-wire PT100 or PT1000. As shown in the figure, wire and adjust the dial switch according to the actual model. And the terminals are PH2.0.

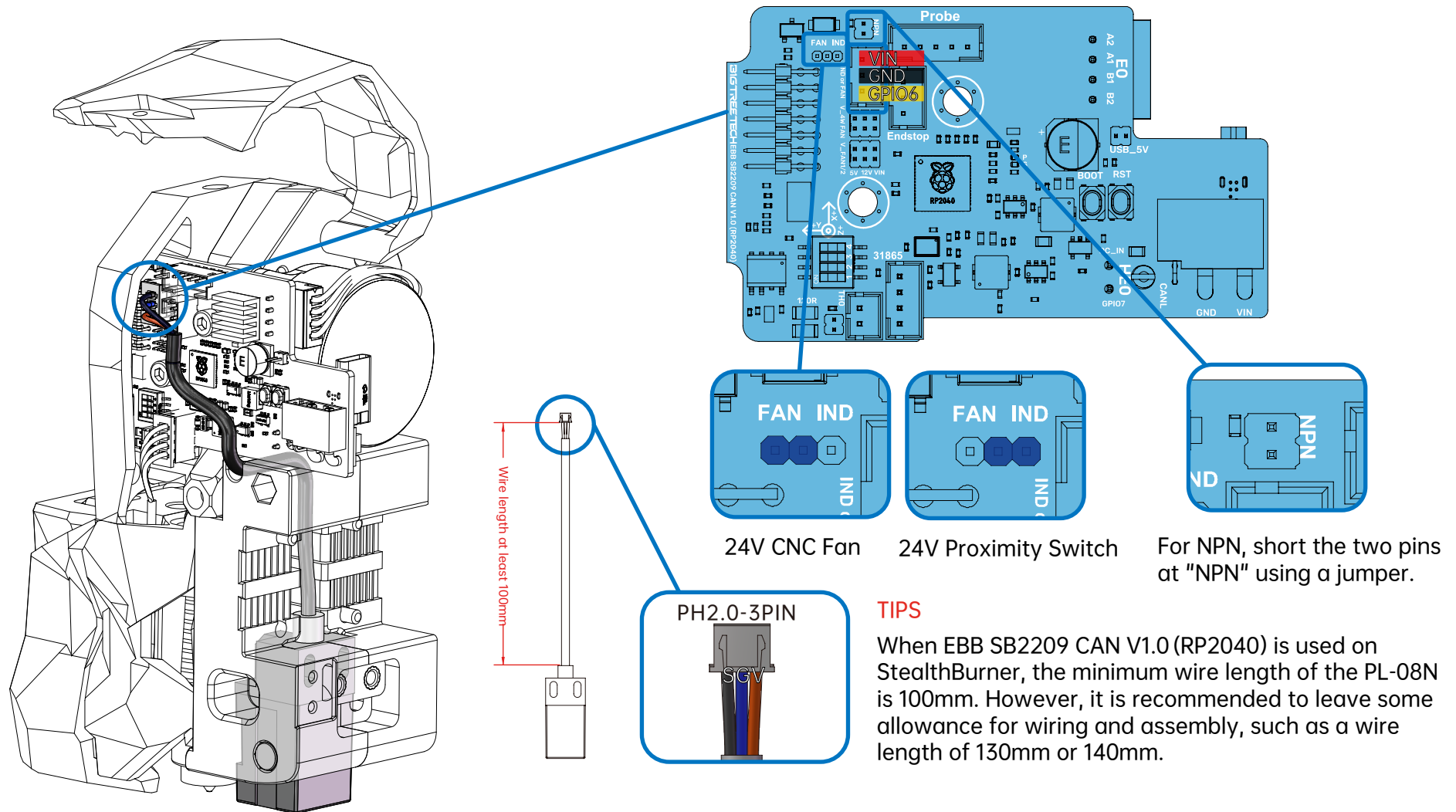


MX3.0-2PIN

TIPS

When EBB SB2209 CAN V1.0 (RP2040) is used on StealthBurner, the minimum wire length of the heater cartridge is 130mm. However, it is recommended to leave some allowance for wiring and assembly, such as a wire length of 150mm or 160mm.

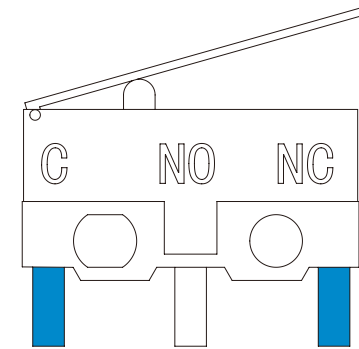
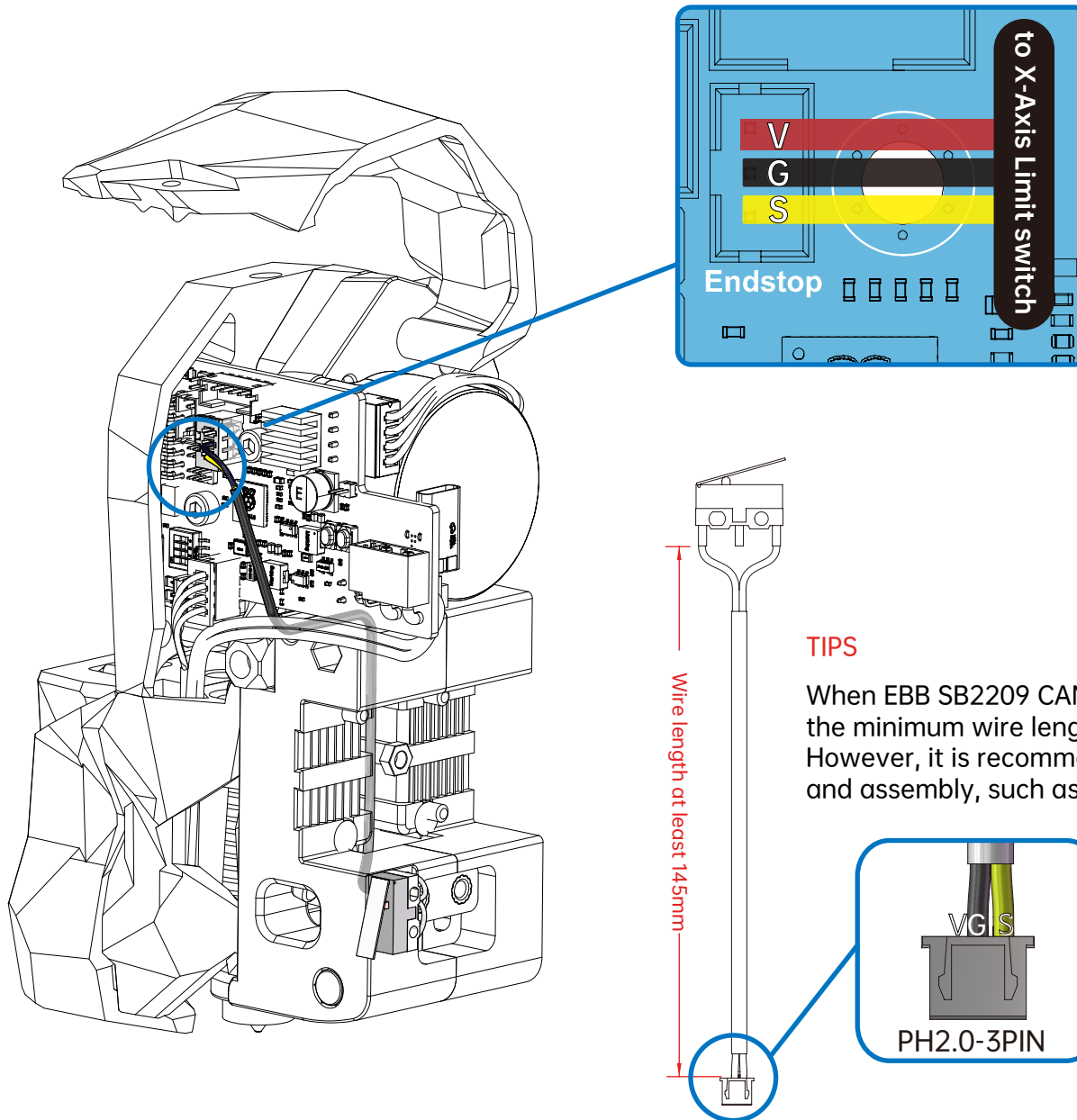
PROXIMITY SWITCH



FUNCTION SELECTION

Supports both NPN and PNP types. For NPN, short the two pins at "NPN" using a jumper; for PNP, remove the jumper.

Note: This interface can also be used for 24V CNC fans. By shorting the left two-pin connector with a jumper (FAN), it can be used for a 24V CNC fan. By shorting the right two-pin connector with a jumper (IND), it can be used for a 24V proximity switch.

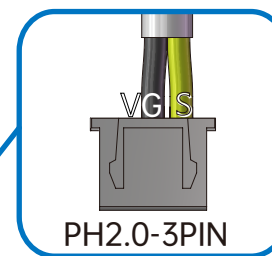


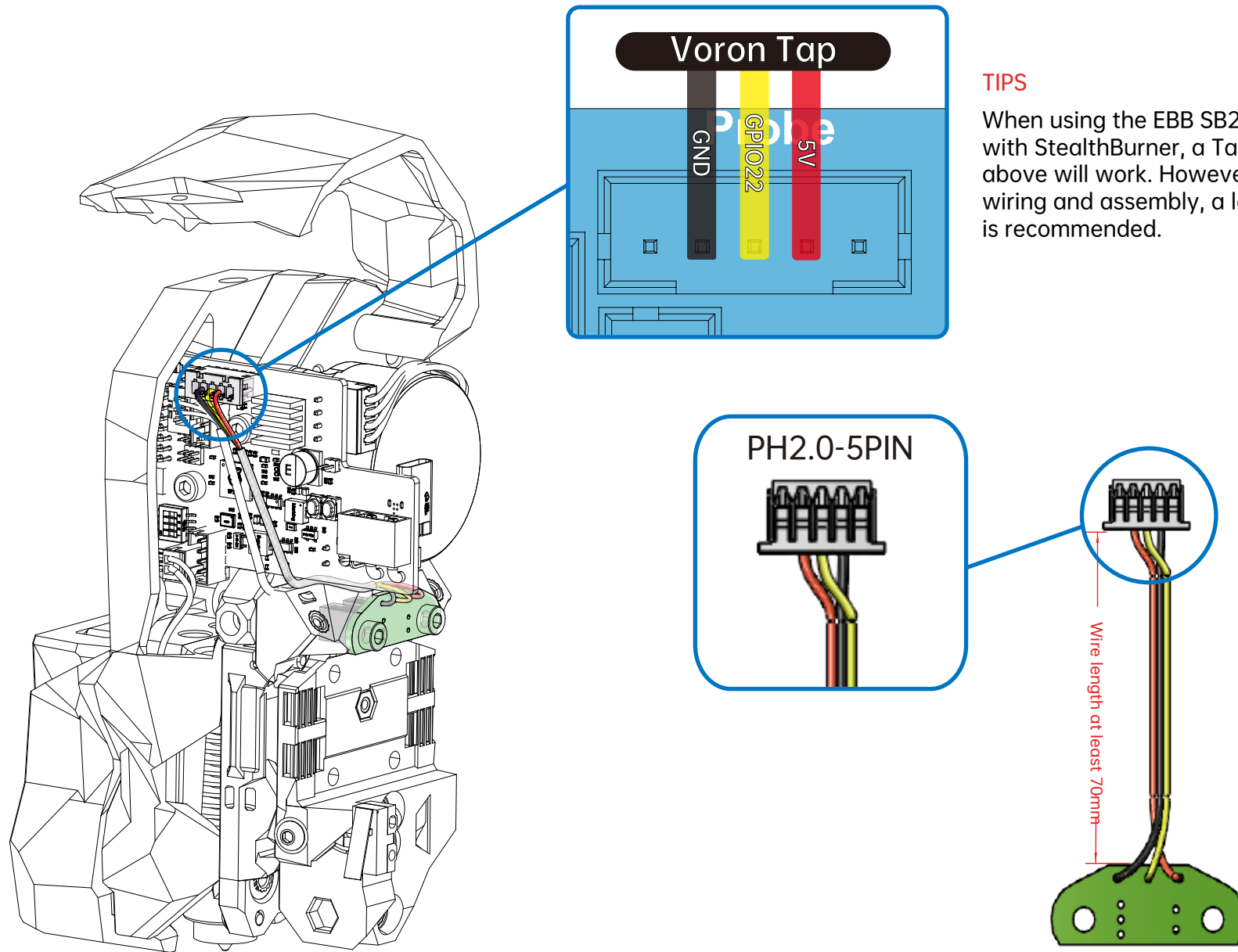
END-STOP SWITCHES FOR X

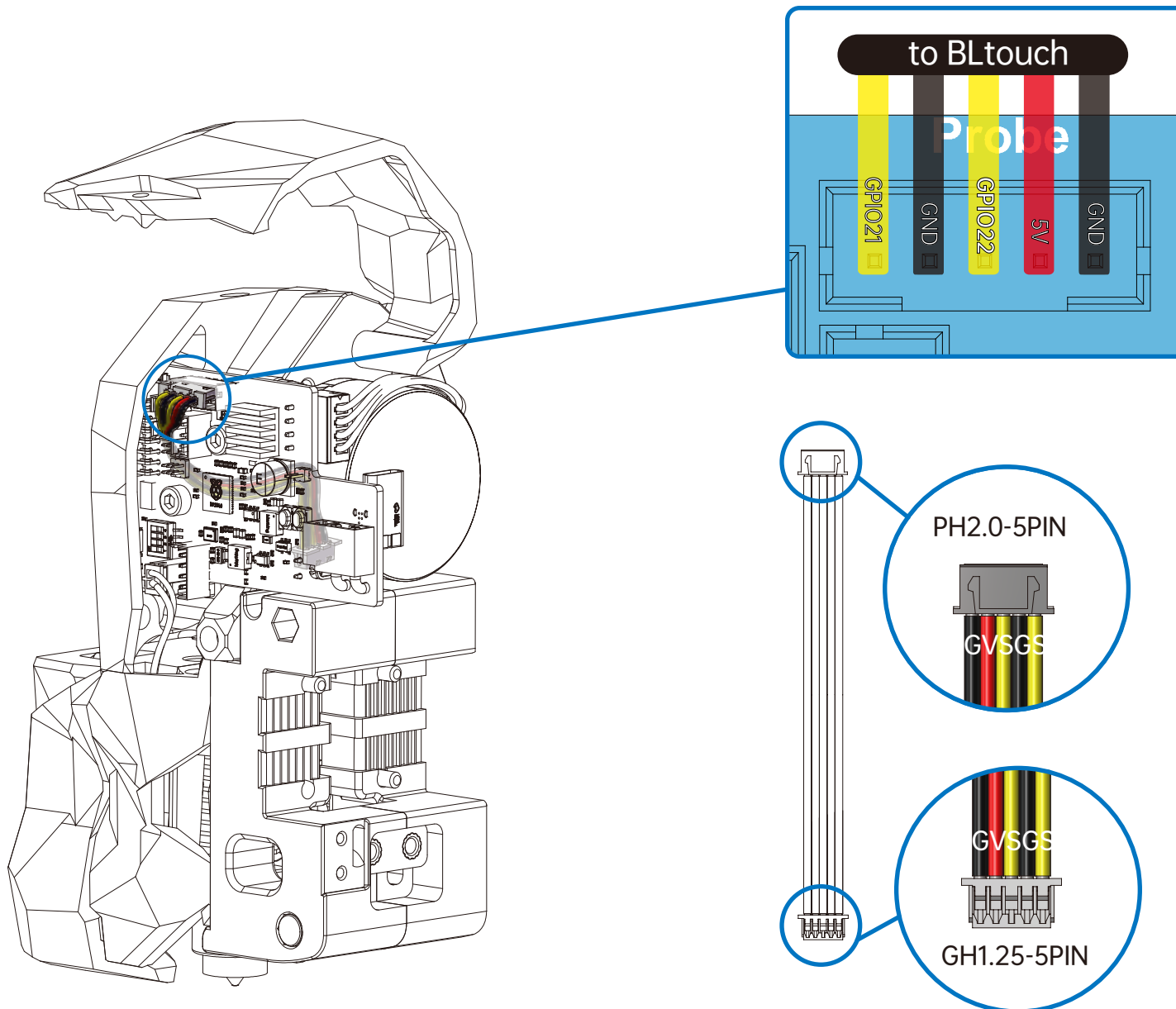
End-stops are wired in a "Normally Closed" configuration. On microswitches those are the 2 outer terminals indicated by C and NC.

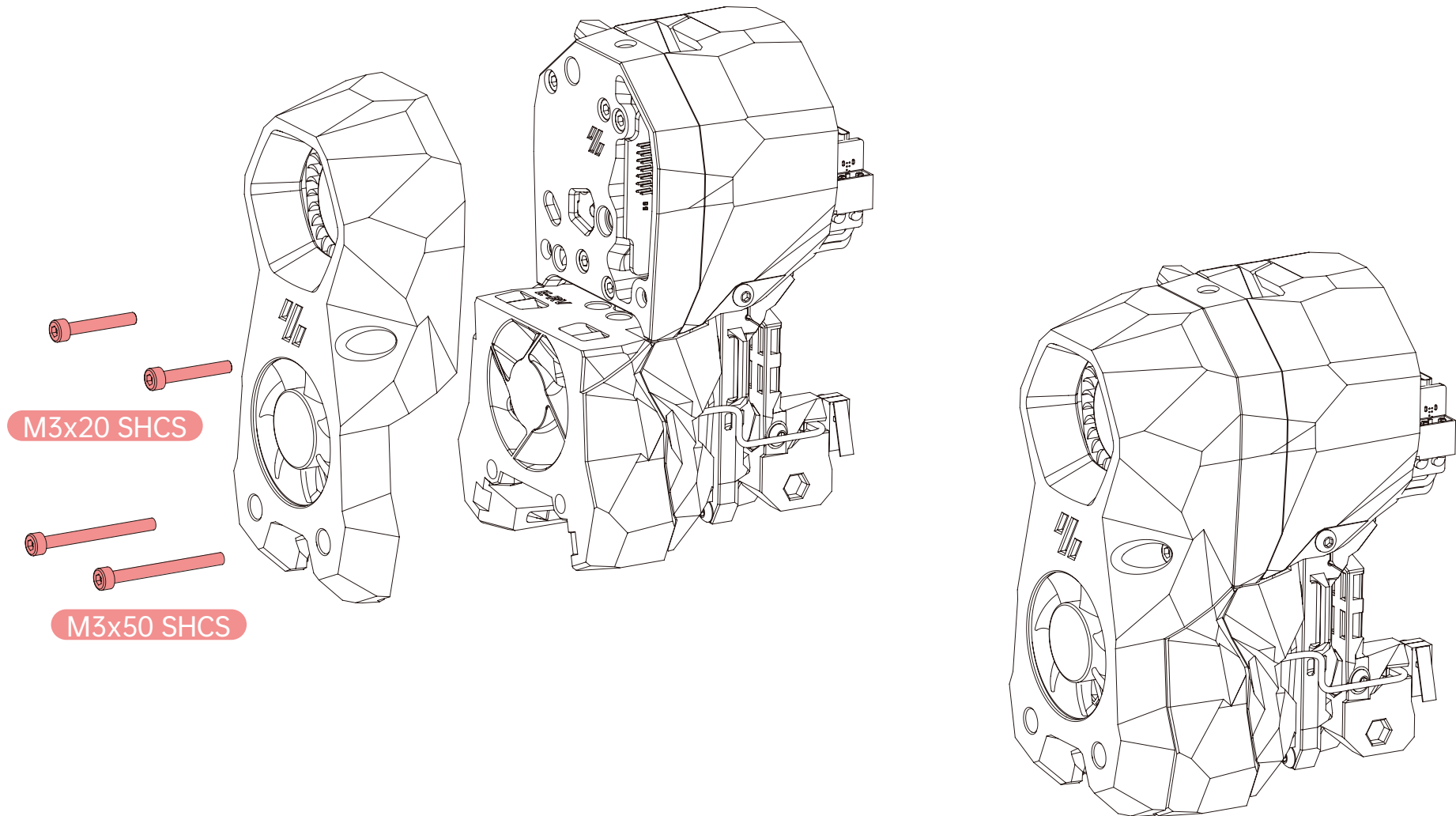
TIPS

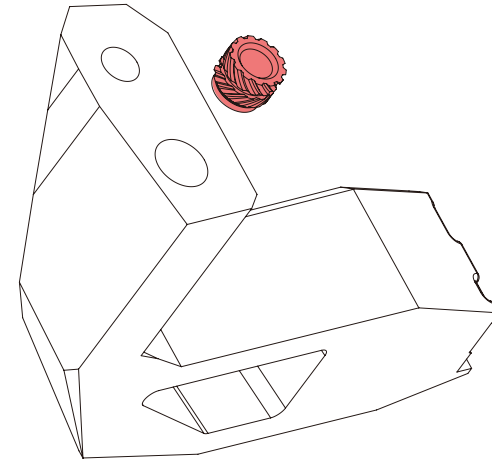
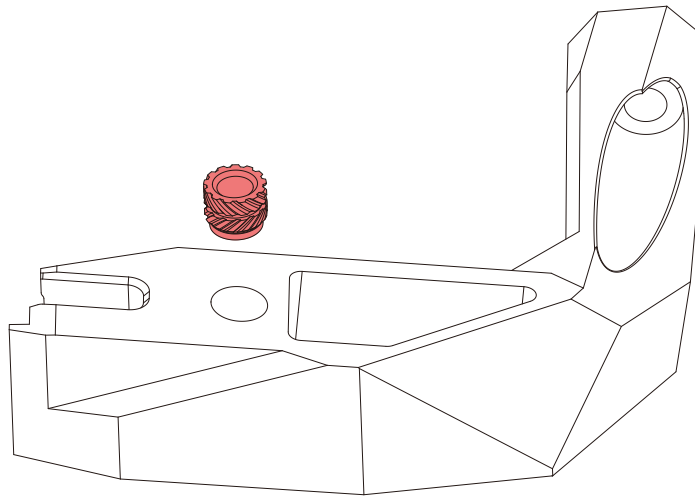
When EBB SB2209 CAN V1.0 (RP2040) is used on StealthBurner, the minimum wire length of the X Axis Limit Switch is 145mm. However, it is recommended to leave some allowance for wiring and assembly, such as a wire length of 165mm or 175mm.

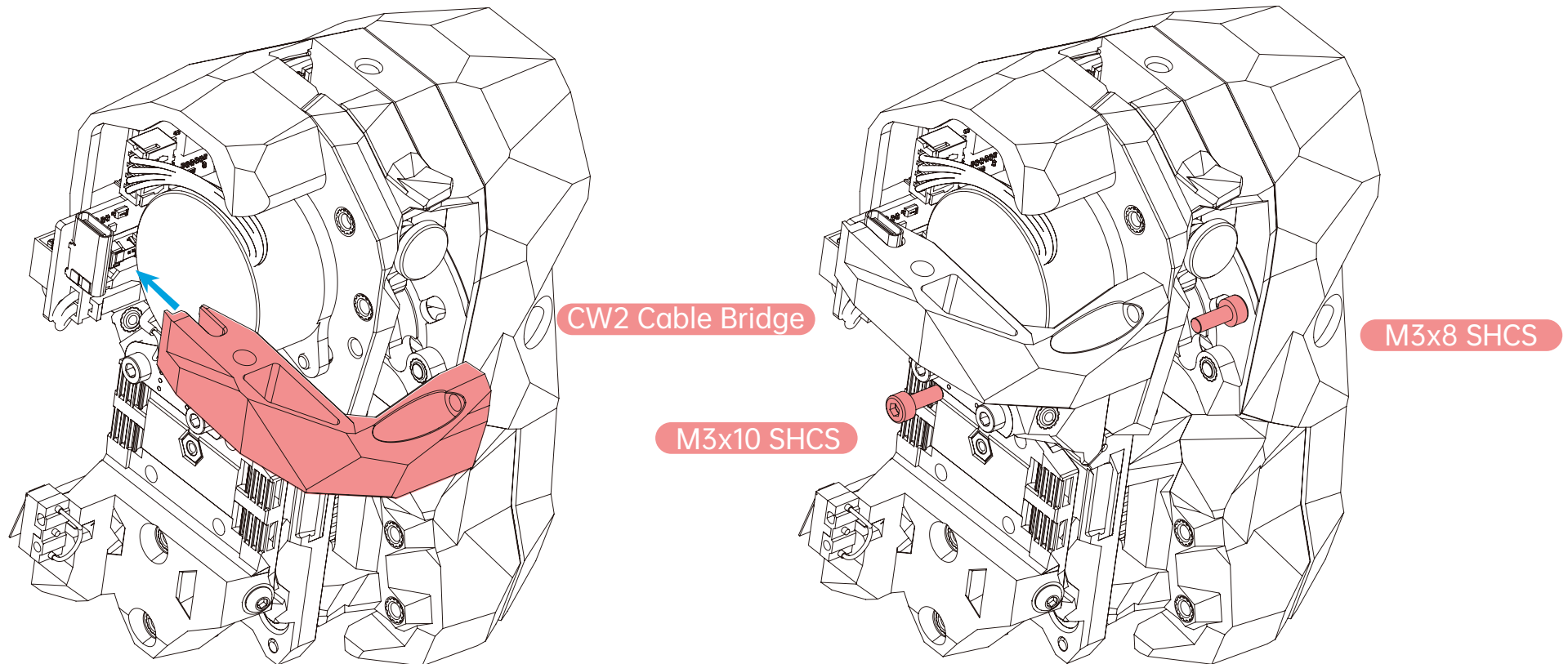






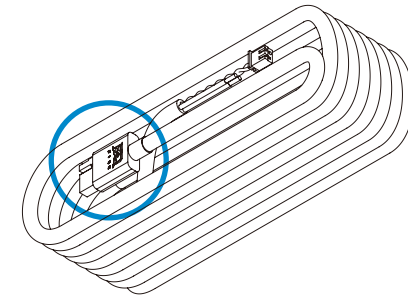
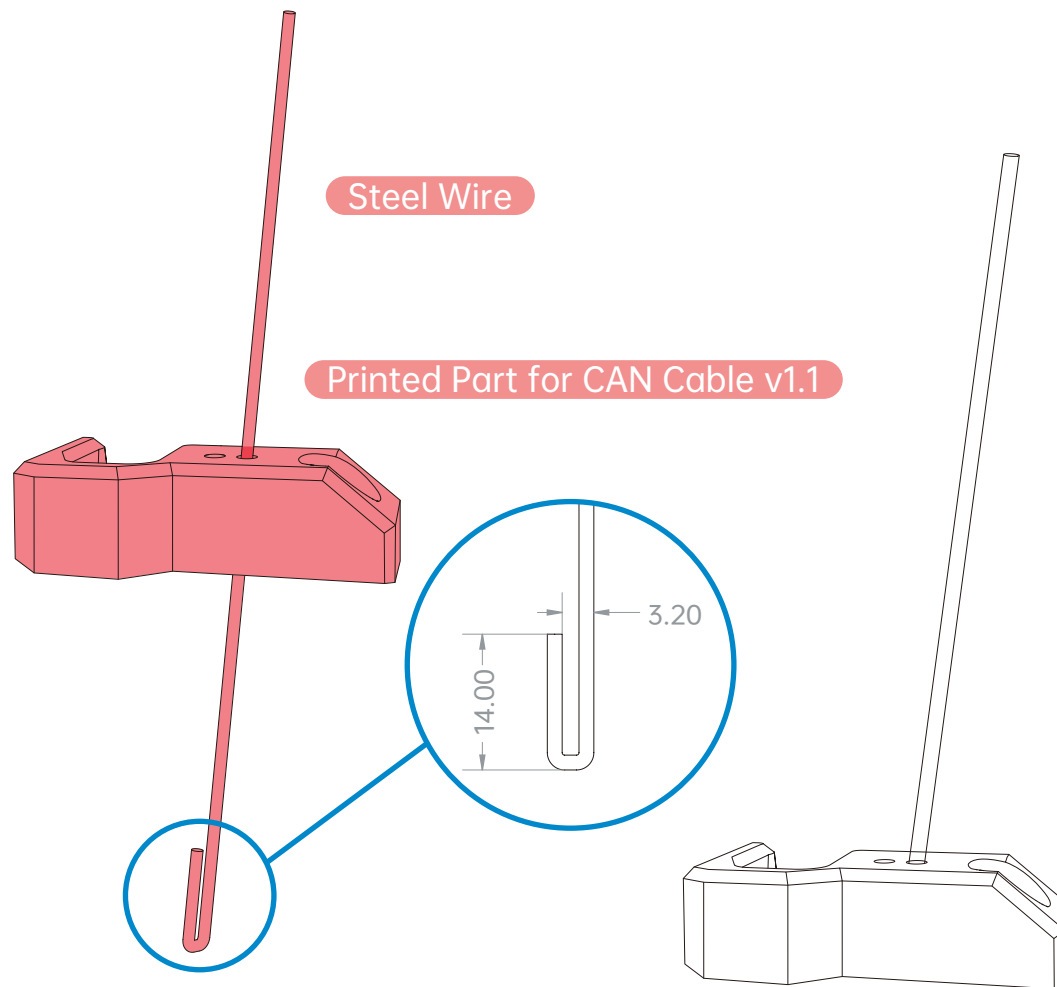






TIPS

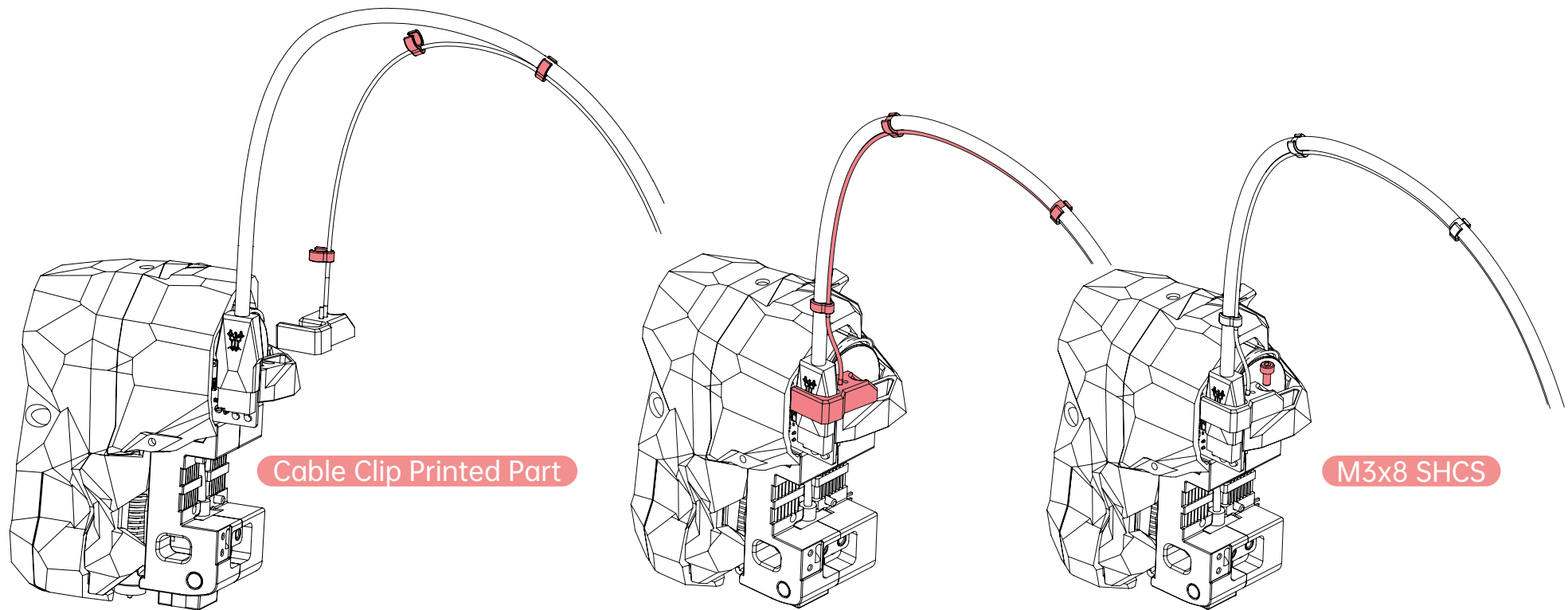
The CW2 cable bridge is different from the official version, it is a custom printed part that can be downloaded directly from our GitHub repository, in EBB\EBB SB2240_2209 CAN\Custom Printed Parts.

**TIPS**

If you are using the previous version of the CAN cable, as shown in the above image, please download the: Printed Part for CAN Cable.

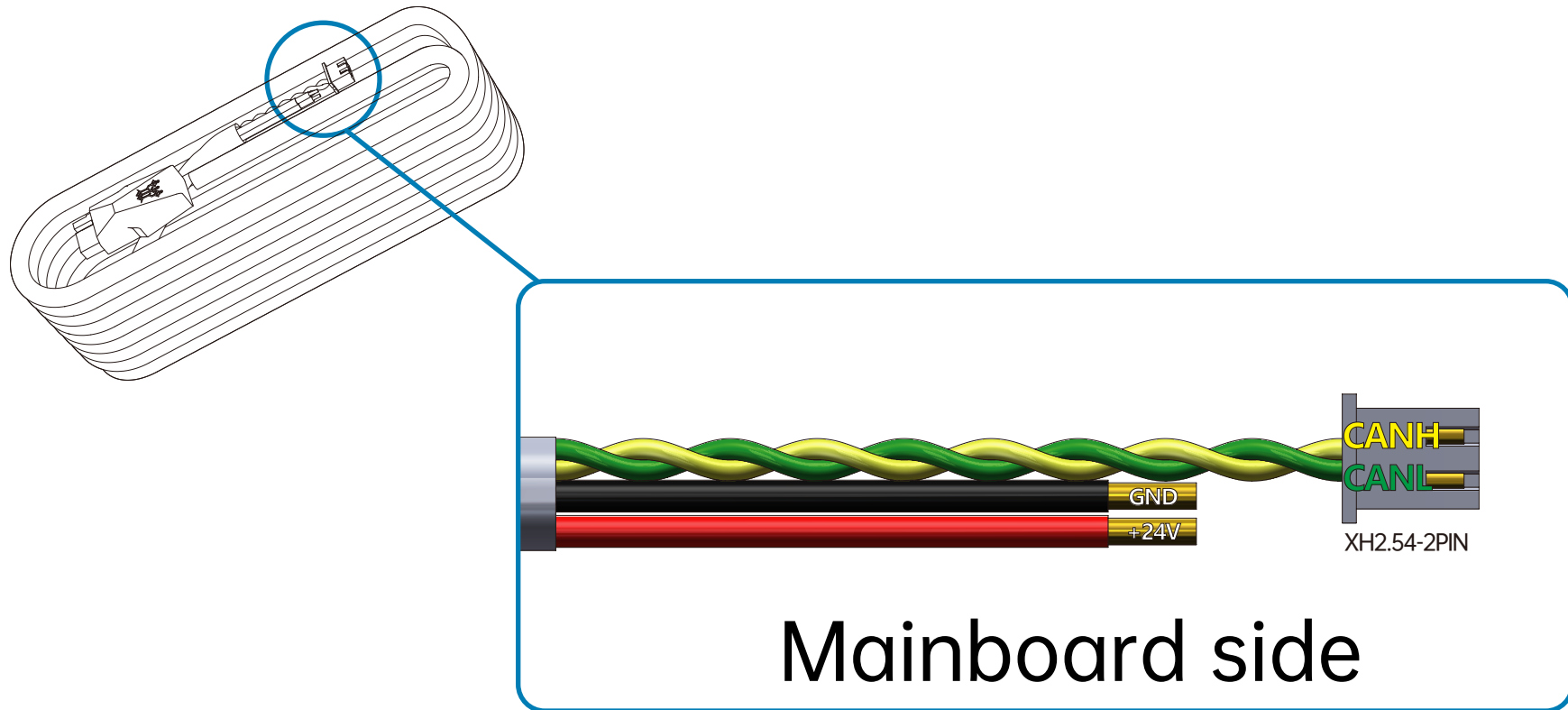
TIPS

The Printed Part for CAN Cable v1.1, Printed Part for USB-C Cable v1.1 are custom printed parts that can be downloaded directly from our GitHub repository, in EBB\EBB SB2240_2209 CAN\Custom Printed Parts.



CABLE CLIP

The cable clip printed part can be downloaded from our GitHub repository, in EBB\EBB SB2240_2209 CAN \Custom Printed Parts, or you can use zip ties instead.



Note: Please note that CanBoot is designed for the purpose of directly updating the MCU firmware via the CAN bus interface. If you prefer the DFU update method, you may skip this step.

"Flashing CanBoot on a CB1/Raspberry Pi"

Refer to the instructions here to download the CanBoot project

<https://github.com/Arksine/CanBoot>

1.Run:

```
cd ~
```

to enter the home directory, then run:

```
git clone https://github.com/Arksine/CanBoot
```

to download CanBoot project.

run:

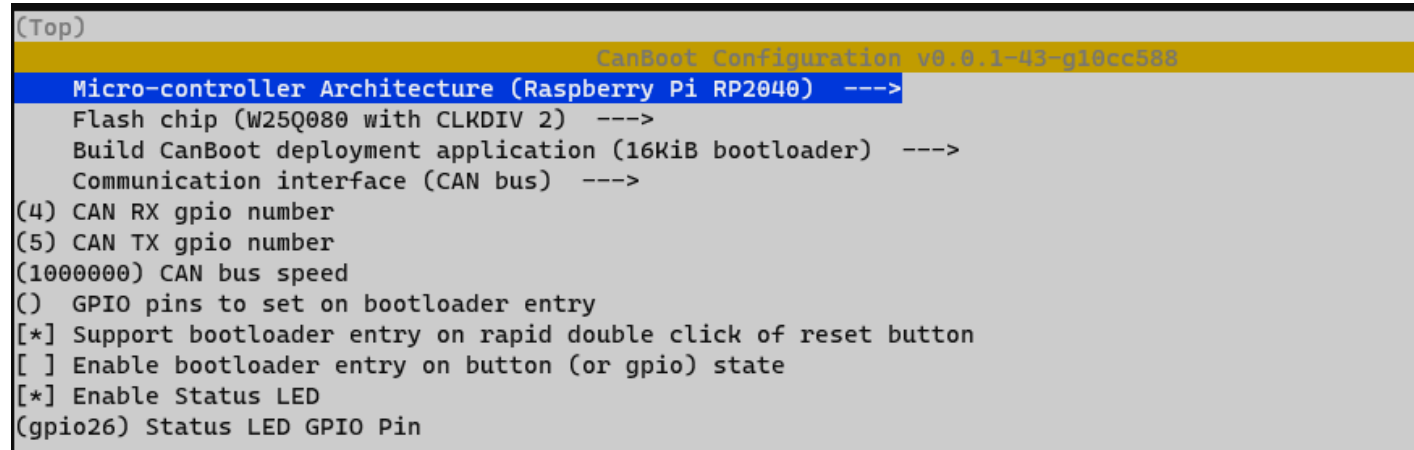
```
cd CanBoot
```

to enter the CanBoot directory.

2.Run:

```
make menuconfig
```

and configure according to the following figure

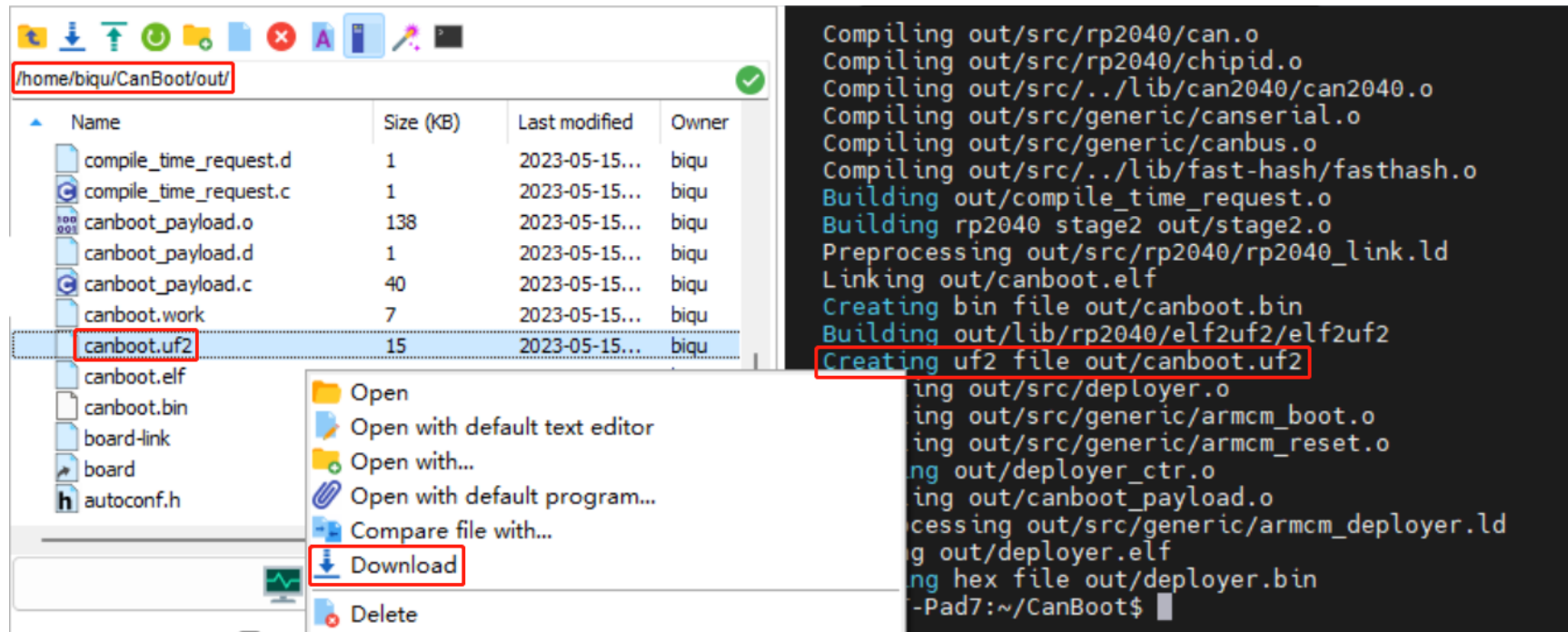


```
(Top)
CanBoot Configuration v0.0.1-43-g10cc588
Micro-controller Architecture (Raspberry Pi RP2040) ---->
Flash chip (W25Q080 with CLKDIV 2) ---->
Build CanBoot deployment application (16KiB bootloader) ---->
Communication interface (CAN bus) ---->
(4) CAN RX gpio number
(5) CAN TX gpio number
(1000000) CAN bus speed
() GPIO pins to set on bootloader entry
[*] Support bootloader entry on rapid double click of reset button
[ ] Enable bootloader entry on button (or gpio) state
[*] Enable Status LED
(gpio26) Status LED GPIO Pin
```

3.Run

`make`

to compile firmware, 'canboot.uf2' file will be generated in `home/biqu/CanBoot/out` folder when `make` is finished, download it onto your computer using the SSH application.

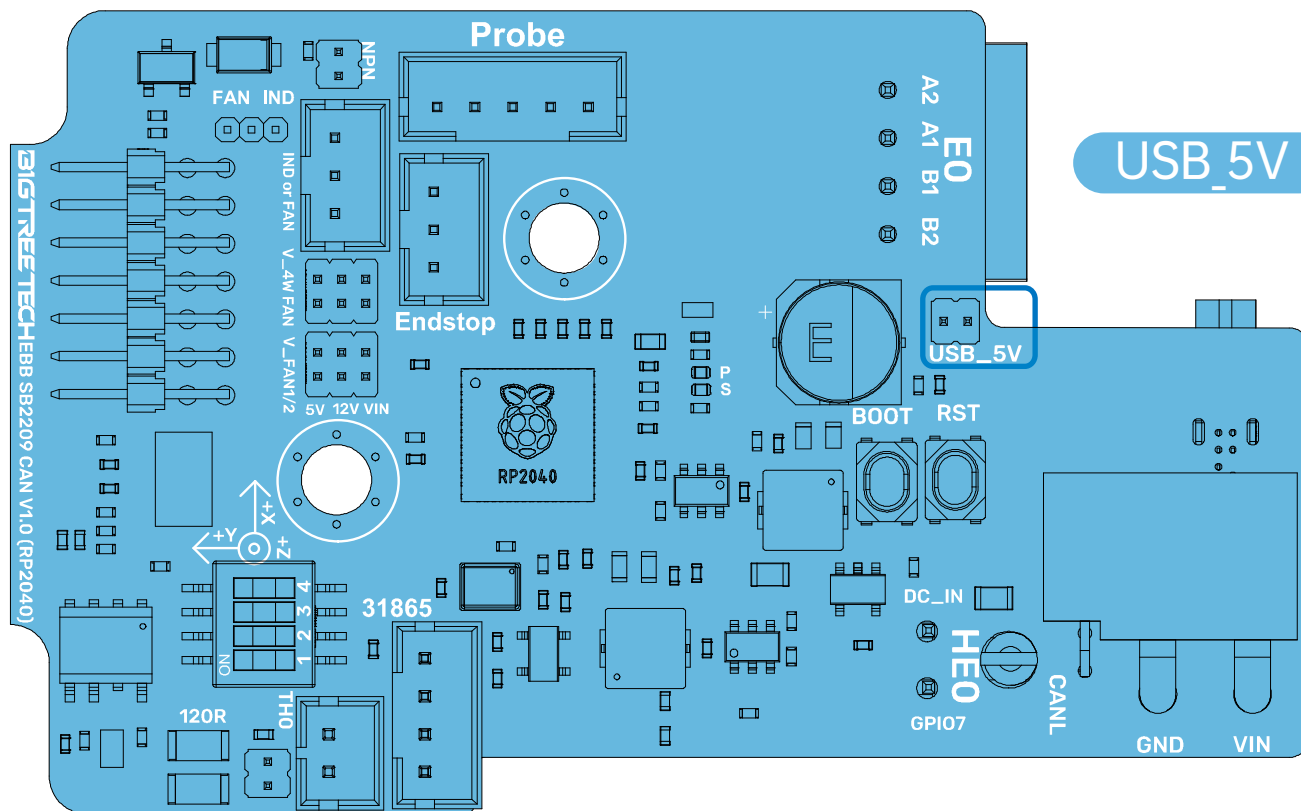


The image shows a file manager window on the left and a terminal window on the right. The file manager displays the contents of the `/home/biqu/CanBoot/out` directory. The file `canboot.uf2` is highlighted, and a context menu is open with the `Download` option selected. The terminal window shows the output of the `make` command, which includes the compilation of various source files and the final creation of the `canboot.uf2` file.

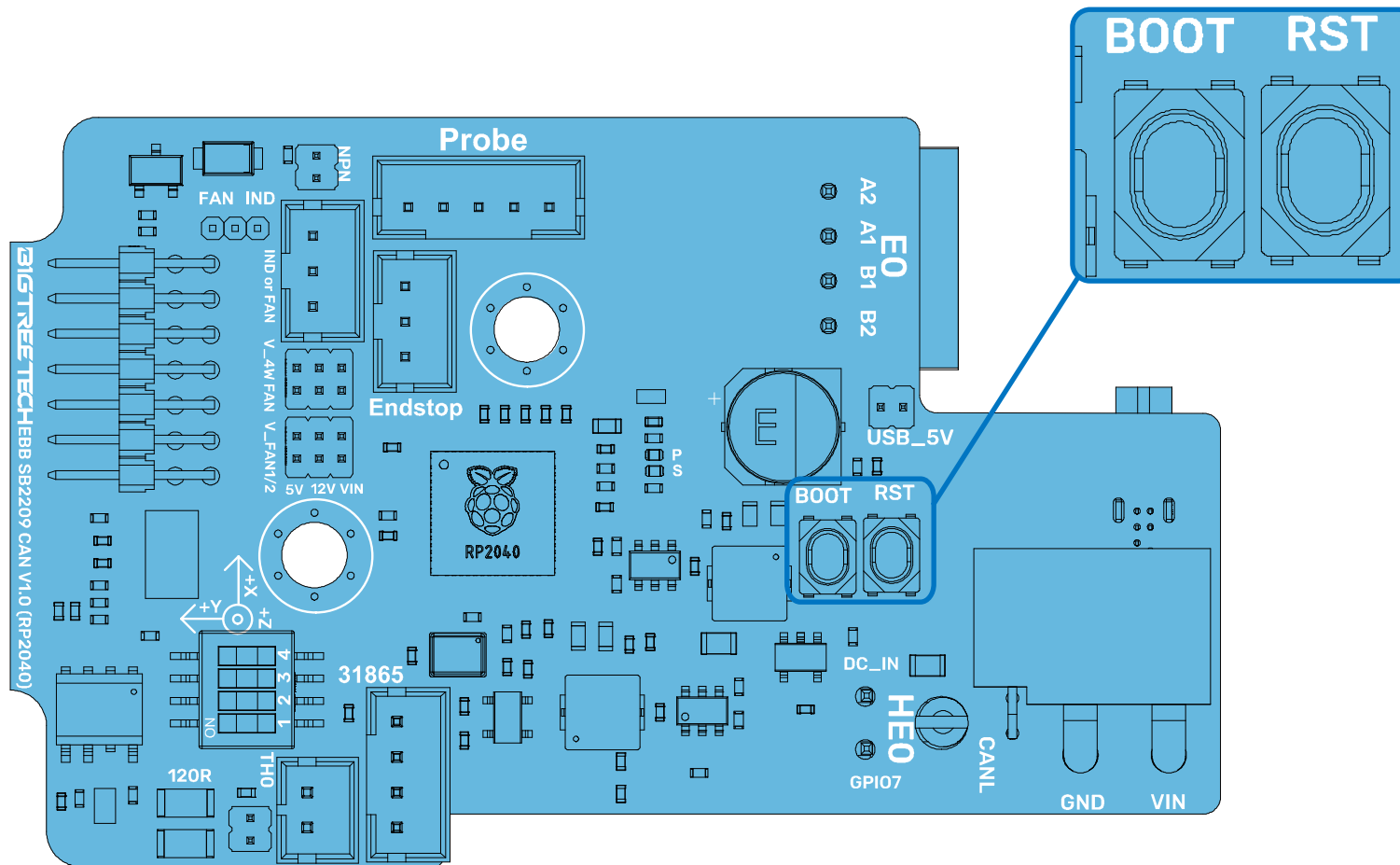
Name	Size (KB)	Last modified	Owner
compile_time_request.d	1	2023-05-15...	biqu
compile_time_request.c	1	2023-05-15...	biqu
canboot_payload.o	138	2023-05-15...	biqu
canboot_payload.d	1	2023-05-15...	biqu
canboot_payload.c	40	2023-05-15...	biqu
canboot.work	7	2023-05-15...	biqu
canboot.uf2	15	2023-05-15...	biqu
canboot.elf			
canboot.bin			
board-link			
board			
autoconf.h			

```
Compiling out/src/rp2040/can.o
Compiling out/src/rp2040/chipid.o
Compiling out/src/./lib/can2040/can2040.o
Compiling out/src/generic/canserial.o
Compiling out/src/generic/canbus.o
Compiling out/src/./lib/fast-hash/fasthash.o
Building out/compile_time_request.o
Building rp2040 stage2 out/stage2.o
Preprocessing out/src/rp2040/rp2040_link.ld
Linking out/canboot.elf
Creating bin file out/canboot.bin
Building out/lib/rp2040/elf2uf2/elf2uf2
Creating uf2 file out/canboot.uf2
Compiling out/src/deployer.o
Compiling out/src/generic/armcm_boot.o
Compiling out/src/generic/armcm_reset.o
Compiling out/deployer_ctr.o
Compiling out/canboot_payload.o
Preprocessing out/src/generic/armcm_deployer.ld
Linking out/deployer.elf
Generating hex file out/deployer.bin
-Pad7:~/CanBoot$
```

4.1 Please use a Type-C cable to connect the EBB SB2209 CAN V1.0 (RP2040) to the Raspberry Pi/CB1, and ensure that the USB_5V jumper is connected, in order to supply power to the EBB SB2209 CAN V1.0 (RP2040) via Type-C.



4.2. Press and hold the Boot button, and then click the Reset button to enter the DFU mode.



4.3. Enter in the SSH terminal command line

```
lsusb
```

Query DFU device ID

```
pi@fluidpi:~$ lsusb
Bus 001 Device 005: ID 2e8a:0003 Raspberry Pi RP2 Boot
Bus 001 Device 004: ID 1d50:6061 OpenMoko, Inc. Geschwister Schneider CAN adapter
Bus 001 Device 003: ID 0424:0c00 Microchip Technology, Inc. (formerly SMSC) SMC9512/9514 Fast Ethernet Adapter
Bus 001 Device 002: ID 0424:9514 Microchip Technology, Inc. (formerly SMSC) SMC9514 Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
pi@fluidpi:~$
```

4.4. Please enter the following command to flash CanBoot:

```
make flash FLASH_DEVICE=2e8a:0003
```

where "2e8a:0003" should be replaced with the actual device ID obtained in step 4.3.

4.5. After flashing is complete, remove the USB_5V jumper and the Type-C cable.

1.) After SSH is successfully connected to Raspberry Pi, run

```
cd ~/klipper/
make menuconfig
```

Compile the firmware with the following configuration (if the options below are not available, please update your Klipper source code to the newest version).

[*] Enable extra low-level configuration options
Micro-controller Architecture (Raspberry Pi RP2040) --->

If you do not use CanBoot

Bootloader offset (No bootloader) --->
Flash chip (W25Q080 with CLKDIV 2) --->

If CanBoot is used

Bootloader offset (16KiB bootloader) --->

If USB communication on Type-C is used

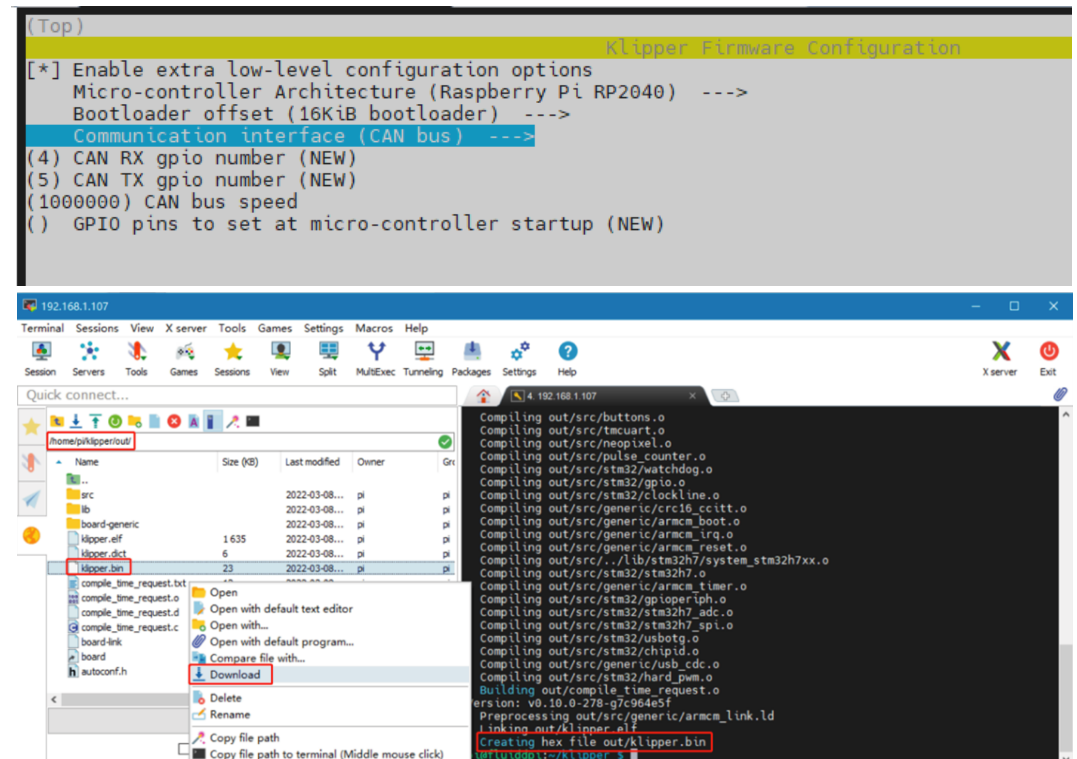
Communication interface (USB) --->

If CAN-Bus communication is used

Communication interface (CAN bus) --->
(4) CAN RX gpio number
(5) CAN TX gpio number
(1000000) CAN bus speed

2.) Press 'q' to exit, and "Yes" when asked to save the configuration.

3.) Run **make** to compile firmware, "klipper. bin" file will be generated in **home/pi/klipper/out** folder when **make** is finished, download it onto your computer using the SSH application.



1.To use the CAN bus, you need to connect the CAN bus cable and insert a jumper at the position of the 120R termination resistor.

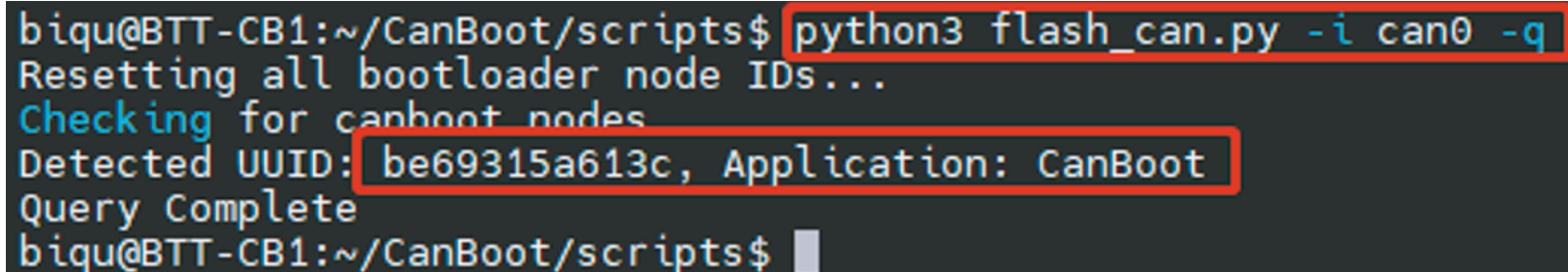
2. Run:

```
cd ~/CanBoot/scripts
```

then run:

```
python3 flash_can.py -i can0 -q
```

to query the canbus ID (connect the CAN cable and power it on in advance), as shown in the figure below, the UUID of the device has been found



```
biqu@BTT-CB1:~/CanBoot/scripts$ python3 flash_can.py -i can0 -q
Resetting all bootloader node IDs...
Checking for canboot nodes
Detected UUID: be69315a613c, Application: CanBoot
Query Complete
biqu@BTT-CB1:~/CanBoot/scripts$
```

3. Run:

```
python3 flash_can.py -i can0 -f ~/klipper/out/klipper.bin -u be69315a613c
```

The be69315a613c is replaced with the actual UUID. Note: klipper.bin needs to be made in advance, and the application start offset of CanBoot is 16KiB offset, so Klipper's menuconfig Bootloader offset should also be 16KiB bootloader, as shown in the following figure.

```
biqu@BTT-CB1:~/CanBoot/scripts$ python3 flash_can.py -i can0 -f ~/klipper/out/klipper.bin -u be69315a613c
Sending bootloader jump command...
Resetting all bootloader node IDs...
Checking for canboot nodes...
Detected UUID: be69315a613c, Application: CanBoot
Attempting to connect to bootloader
CanBoot Connected
Protocol Version: 1.0.0
Block Size: 64 bytes
Application Start: 0x8002000
MCU type: stm32g0b1xx
Verifying canbus connection
Flashing '/home/biqu/klipper/out/klipper.bin'...

[#####]

Write complete: 13 pages
Verifying (block count = 414)...

[#####]

Verification Complete: SHA = C3B1F96A8FCE706587BF4A9119D95D80465875A3
CAN Flash Success
biqu@BTT-CB1:~/CanBoot/scripts$
```

4. Run:

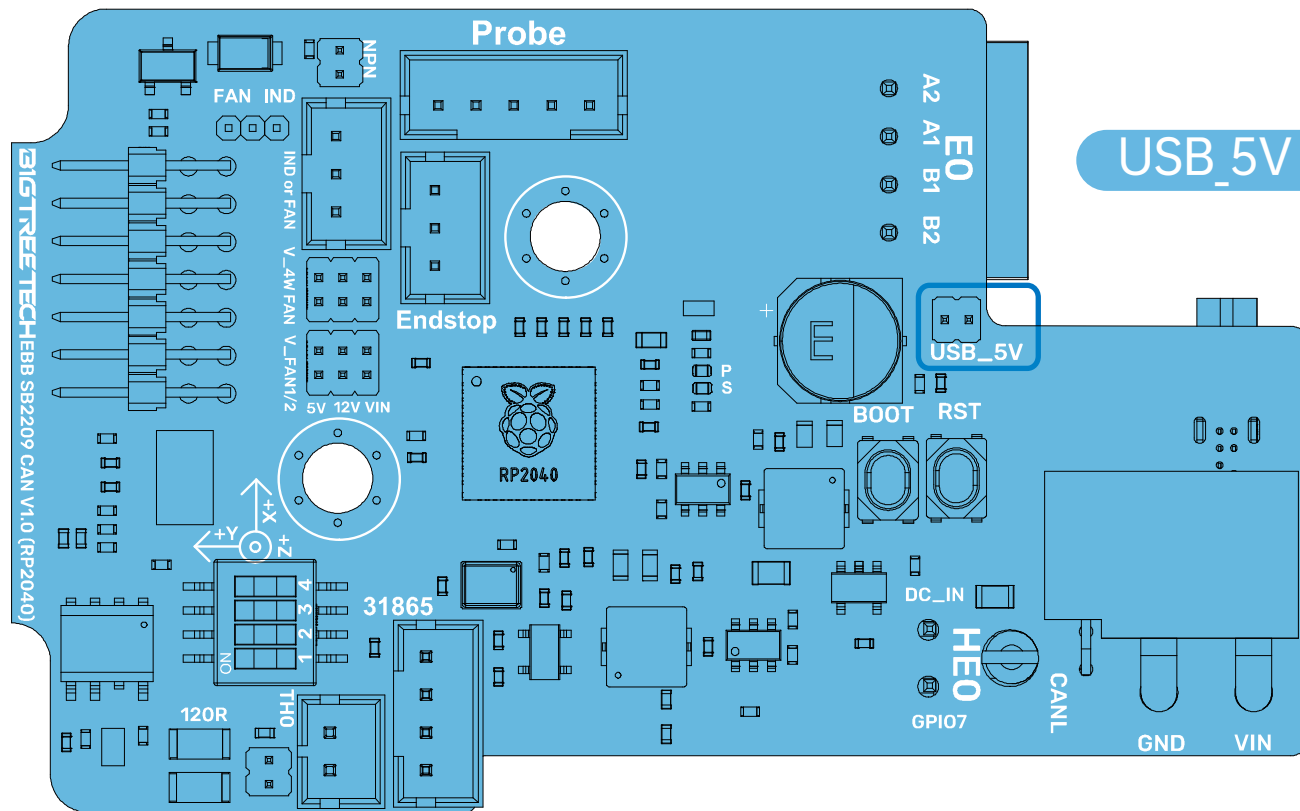
```
python3 flash_can.py -i can0 -q
```

to query. At this time, the Application changes from CanBoot to Klipper, indicating that Klipper has been running normally

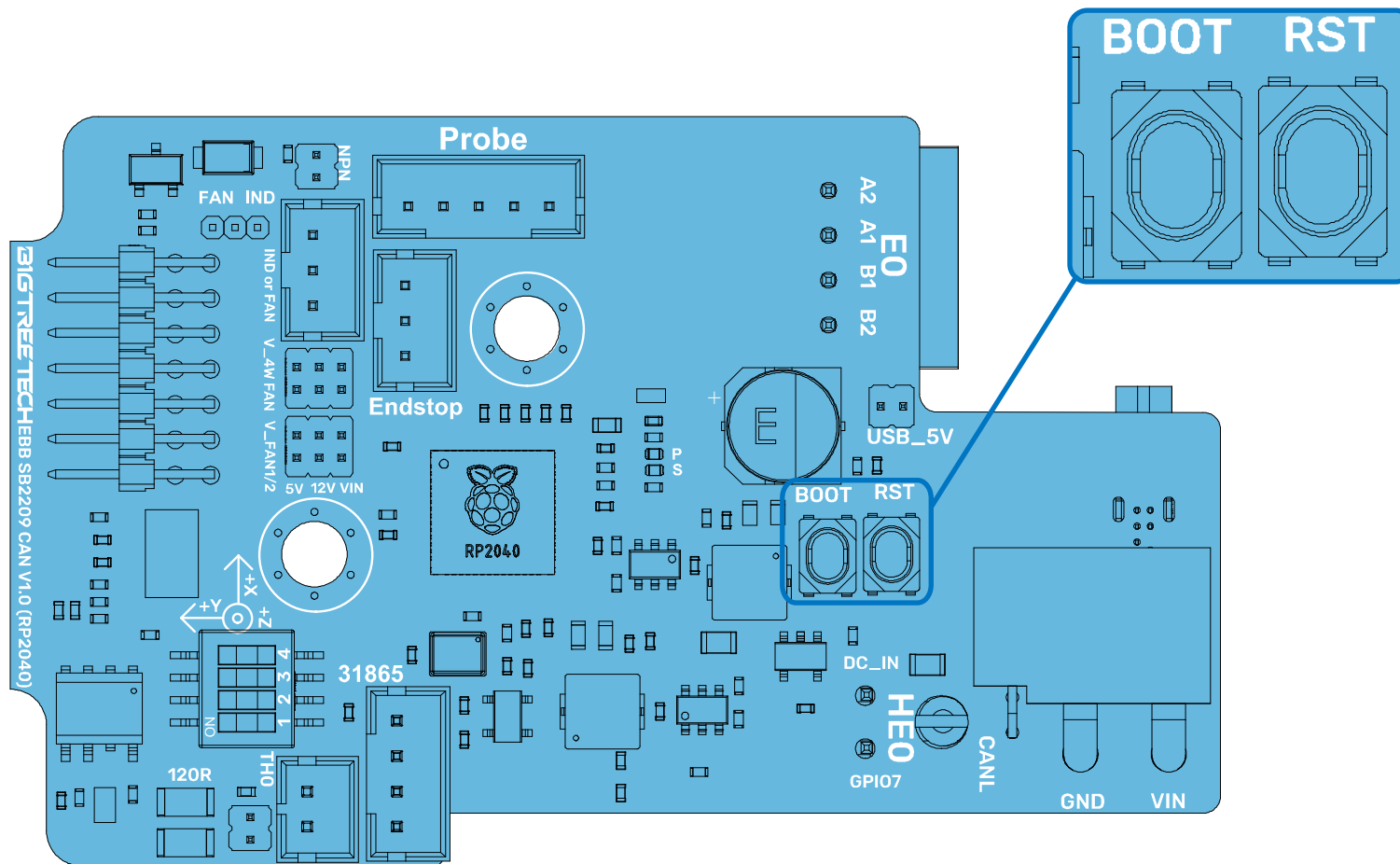
```
biqu@BTT-CB1:~/CanBoot/scripts$ python3 flash_can.py -i can0 -q
Resetting all bootloader node IDs...
Checking for canboot nodes...
Detected UUID: be69315a613c, Application: Klipper
Query Complete
biqu@BTT-CB1:~/CanBoot/scripts$
```

Raspberry Pi or CB1 update via DFU.

1. Please use a Type-C cable to connect the EBB SB2209 CAN V1.0 (RP2040) to the Raspberry Pi/CB1, and ensure that the USB_5V jumper is connected, in order to supply power to the EBB SB2209 CAN V1.0 (RP2040) via Type-C.



2. Press and hold the Boot button, and then click the Reset button to enter the DFU mode.



3. Enter in the SSH terminal command line

```
lsusb
```

Query DFU device ID

```
pi@fluidpi:~$ lsusb
Bus 001 Device 005: ID 2e8a:0003 Raspberry Pi RP2 Boot
Bus 001 Device 004: ID 1d50:6061 OpenMoko, Inc. Geschwister Schneider CAN adapter
Bus 001 Device 003: ID 0424:0c00 Microchip Technology, Inc. ( formerly SMSC ) SMC9512/9514 Fast Ethernet Adapter

Bus 001 Device 002: ID 0424:9514 Microchip Technology, Inc. ( formerly SMSC ) SMC9514 Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
pi@fluidpi:~$
```

4. Run:

```
cd klipper
```

to enter to the klipper directory, then run the following command to write the firmware:

```
make flash FLASH_DEVICE=2e8a:0003
```

Note: Replace 2e8a:0003 with the actual device ID found in the previous step.

5. After the firmware is written, run

```
ls /dev/serial/by-id/
```

to query the serial ID of the device (this ID can only be found in the USB communication mode, and this step is ignored in CANBus mode).

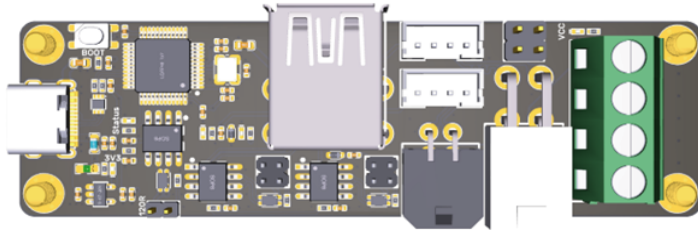
6. If USB communication is used, after the first writing, it is not necessary to manually press the Boot and Reset buttons to enter the DFU mode when updating again. You can directly enter

```
make flash FLASH_DEVICE=/dev/serial/by-id/usb-Klipper_rp2040_4550357128922FC8-if00
```

to write the firmware (Note: replace `/dev/serial/by id/xxx` with the actual ID found in the previous step).

After flashing is complete, remove the USB_5V jumper and the Type-C cable.

Used with BIGTREETECH U2C module.



1. Enter the command

```
sudo nano /etc/network/interfaces.d/can0
```

in the SSH terminal and execute

```
allow-hotplug can0
iface can0 can static
    bitrate 1000000
up ifconfig $IFACE txqueuelen 1024
```

Set the CAN-BUS speed to 1M (it must be consistent with the speed set in the firmware (1000000) CAN bus speed), save (Ctrl+S) and exit (Ctrl+X) after modification, and enter

```
sudo reboot
```

to restart Raspberry Pi.

2. Each device on CAN bus will generate a `canbus_uuid` according to the UID of MCU, to find each microcontroller device ID, make sure the hardware is powered on and wired correctly, and then run:

```
~/klippy-env/bin/python ~/klipper/scripts/canbus_query.py can0
```

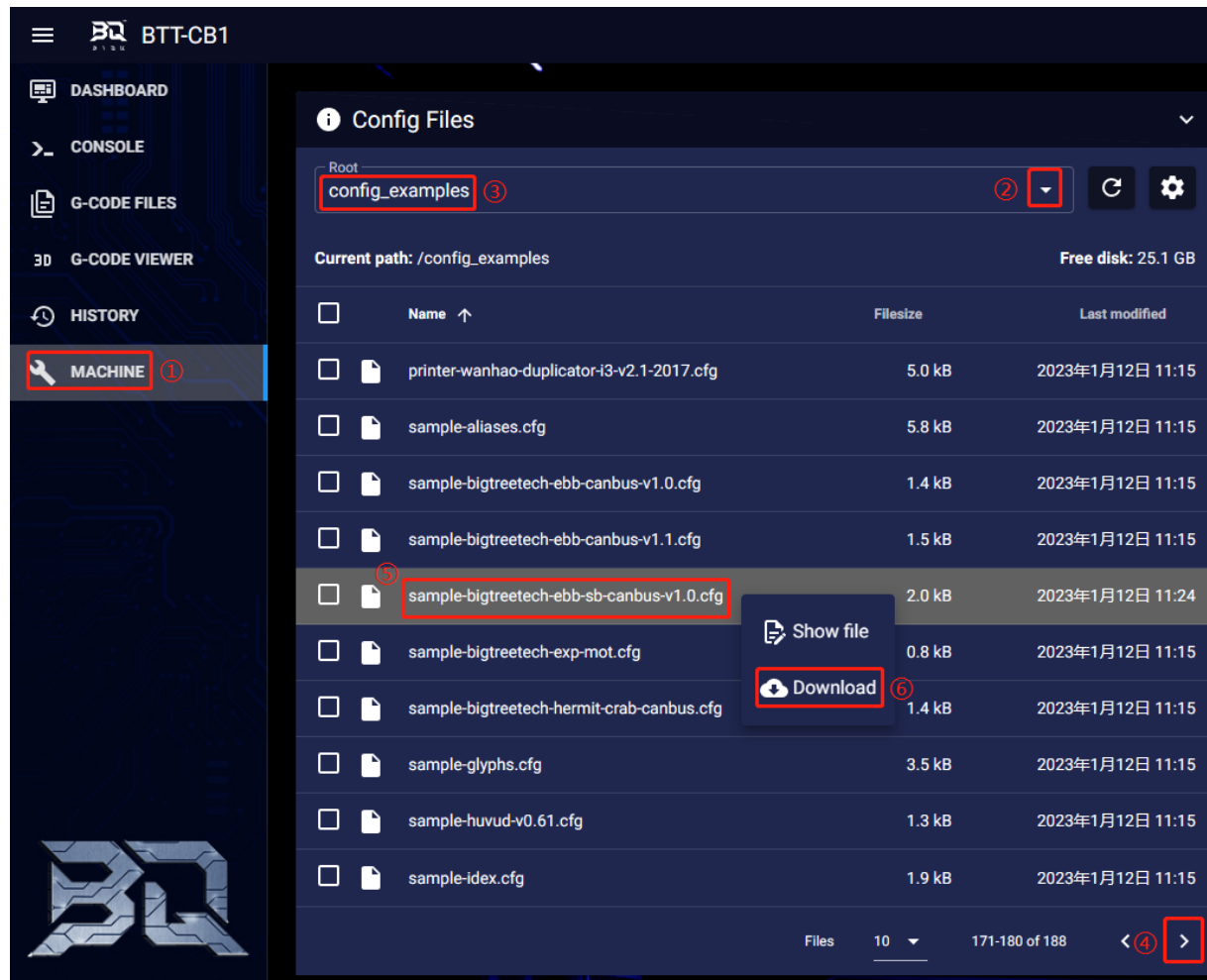
3. If an uninitialized CAN device is detected, the above command will report the device's `canbus_uuid`.

```
Found canbus_uuid=0e0d81e4210c
```

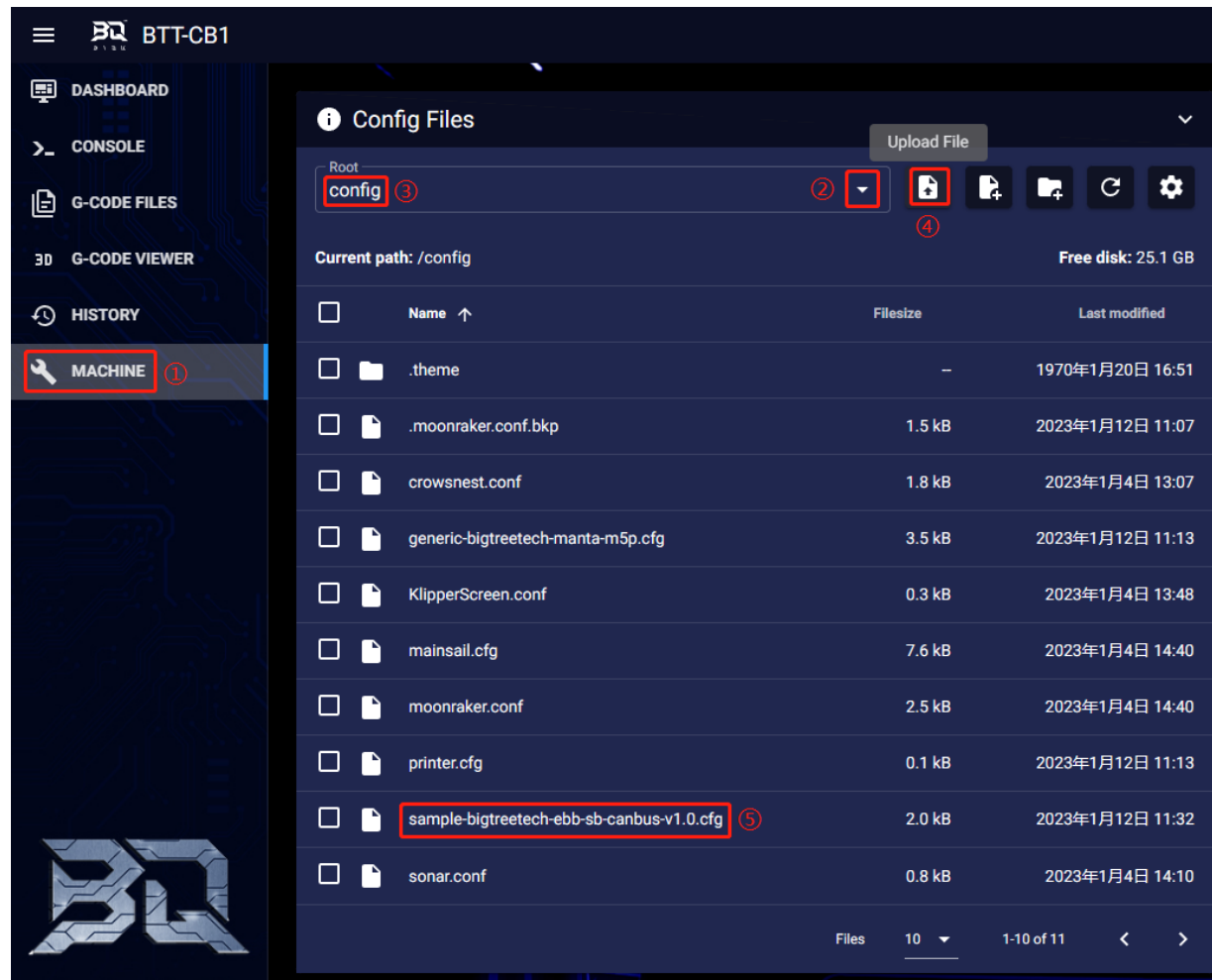
4. If Klipper has been running normally and connected to this device, then `canbus_uuid` will not be reported, which is normal.

1. Enter Raspberry Pi IP address into your browser, and find the reference config for the motherboard in the directory shown below, if there is no such config available, update your Klipper source code to the newest version or download it from GitHub:

<https://github.com/bigtreetech/EBB>



2. Upload the configuration file of the motherboard to Configuration Files.



3. Add the configuration of this motherboard in the "printer.cfg" file:

```
[include sample-bigtreetech-ebb-sb-canbus-v1.0.cfg]
```

X printer.cfg

11

12

13

```
[include sample-bigtreetech-ebb-canbus-v1.0.cfg]
```

4. Enter the correct ID (USB serial or canbus).

X sample-bigtreetech-ebb-sb-canbus-v1.0.cfg

8

9

10

```
[mcu EBBCan]
```

```
serial: /dev/serial/by-id/usb-Klipper_firmware_12345-if00
```

```
#canbus_uuid: 0e0d81e4210c
```

5. Configure the module's specific functions according to

<https://www.klipper3d.org/Overview.html>

Website

www.bigtree-tech.com

GitHub

www.github.com/bigtreetech

Discord

www.discord.gg/5jdwbYYZuv

